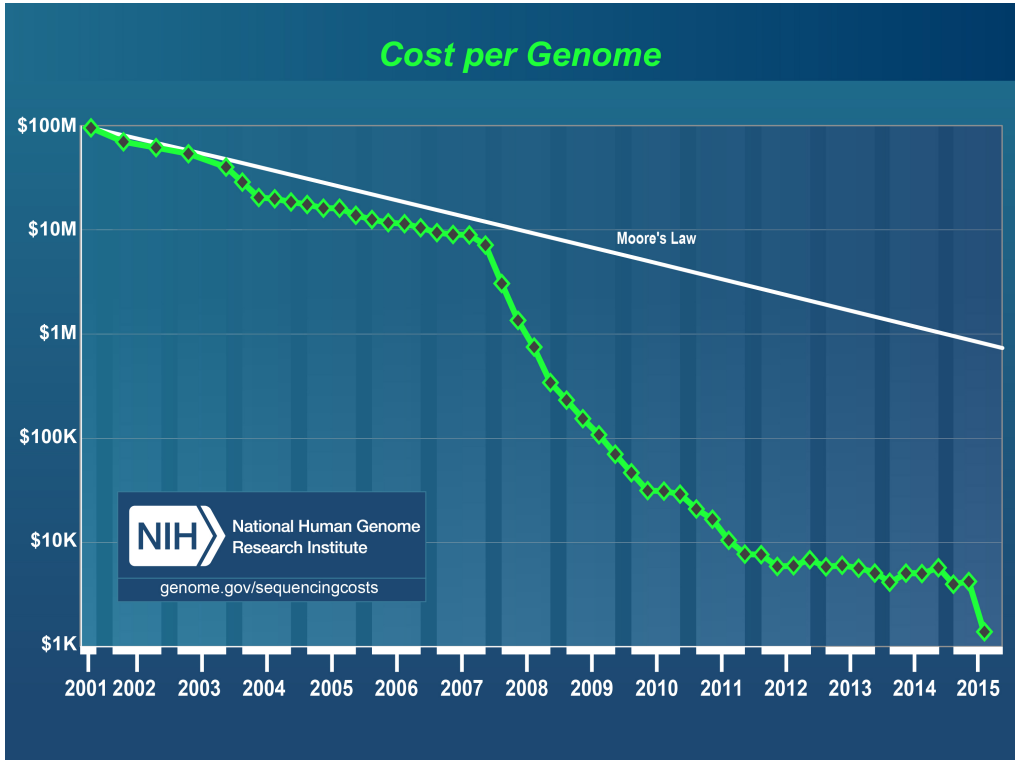
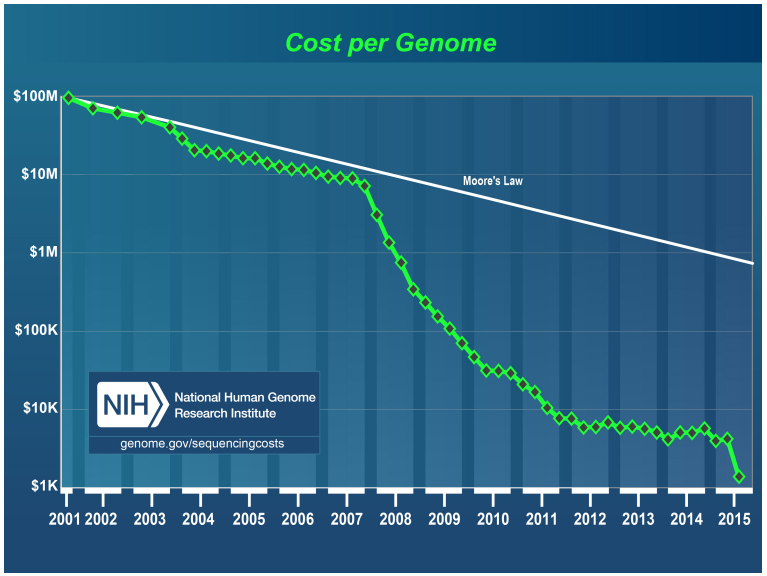
Lecture 3: *Sequencing*



Shotgun, new sequencing technology, nanopores

Outline

1. **Sanger Sequencing**
2. Next Generation Sequencers
3. Third Generation Sequencers



Sequencing Revolution

- Advances in our ability to sequence DNA lies at the heart of the biological revolution
- Despite the structure of DNA being discovered in 1953, reading DNA sequences took several decades to take off
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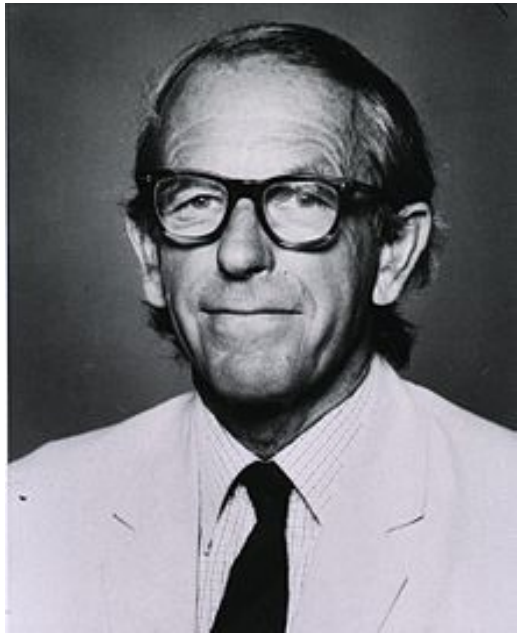

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- In 1955 he discovered the protein sequence for Insulin
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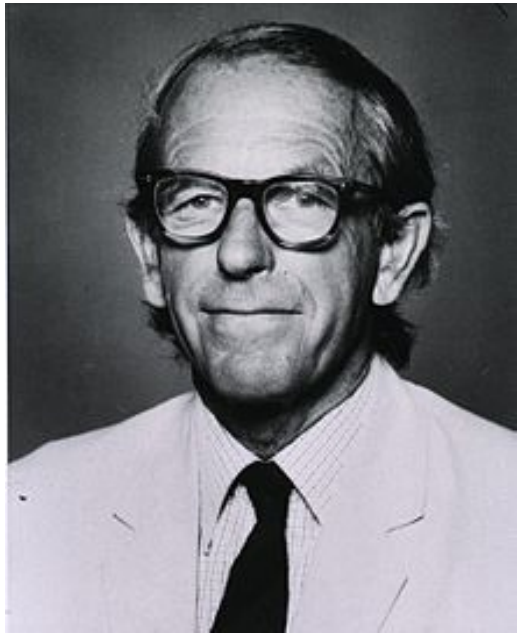

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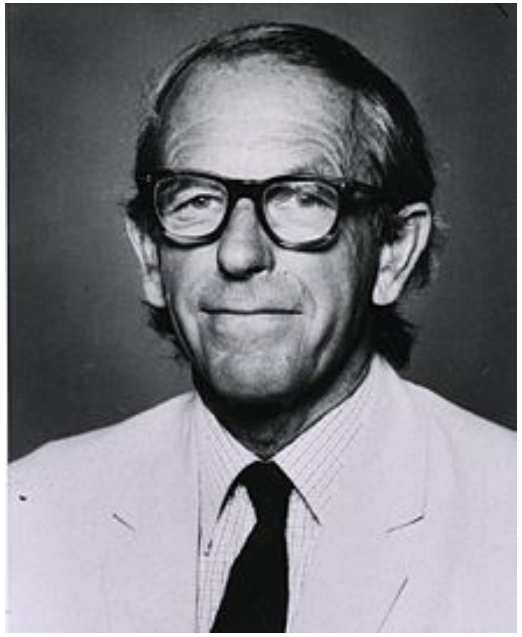

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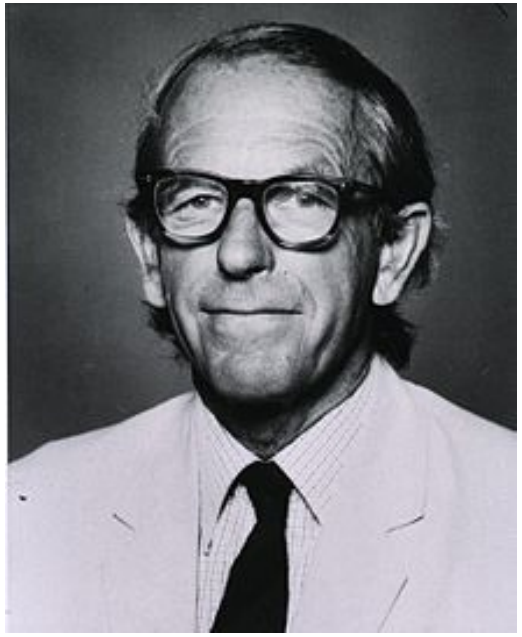

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- It wasn't until 1965 that the very first strands of RNA were read
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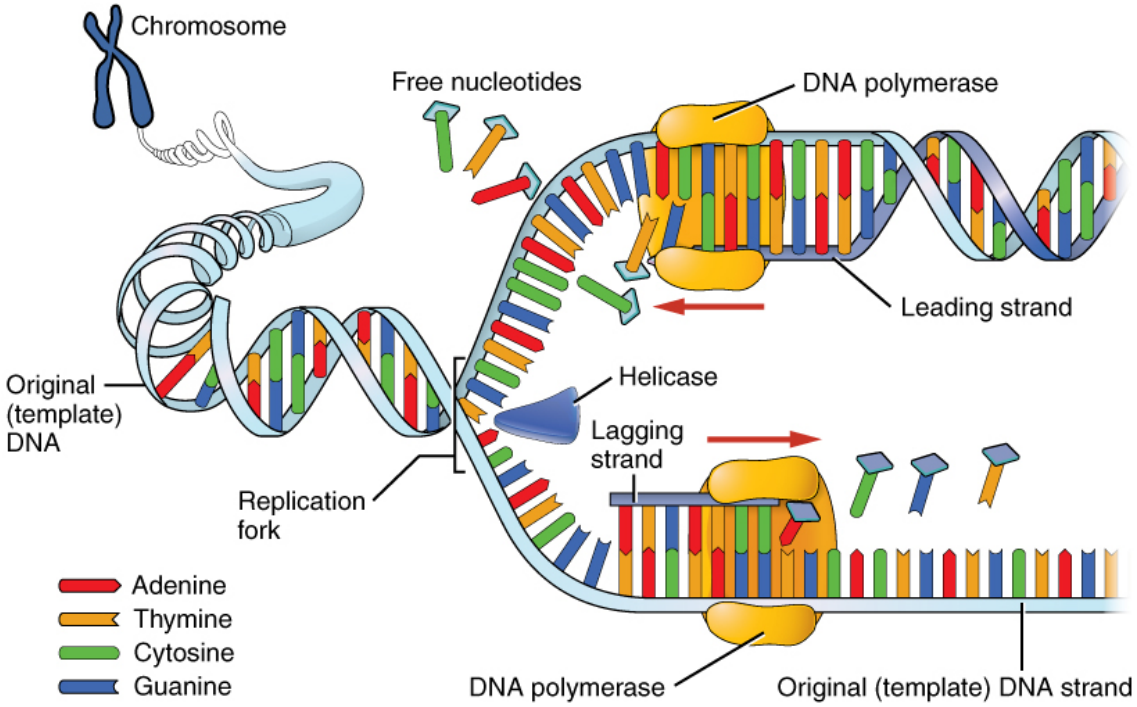
Sanger Sequencing

- In Sanger sequencing we use biological mechanisms for copying DNA
- This would add Dideoxynucleoties (ddNTP) to a DNA chain



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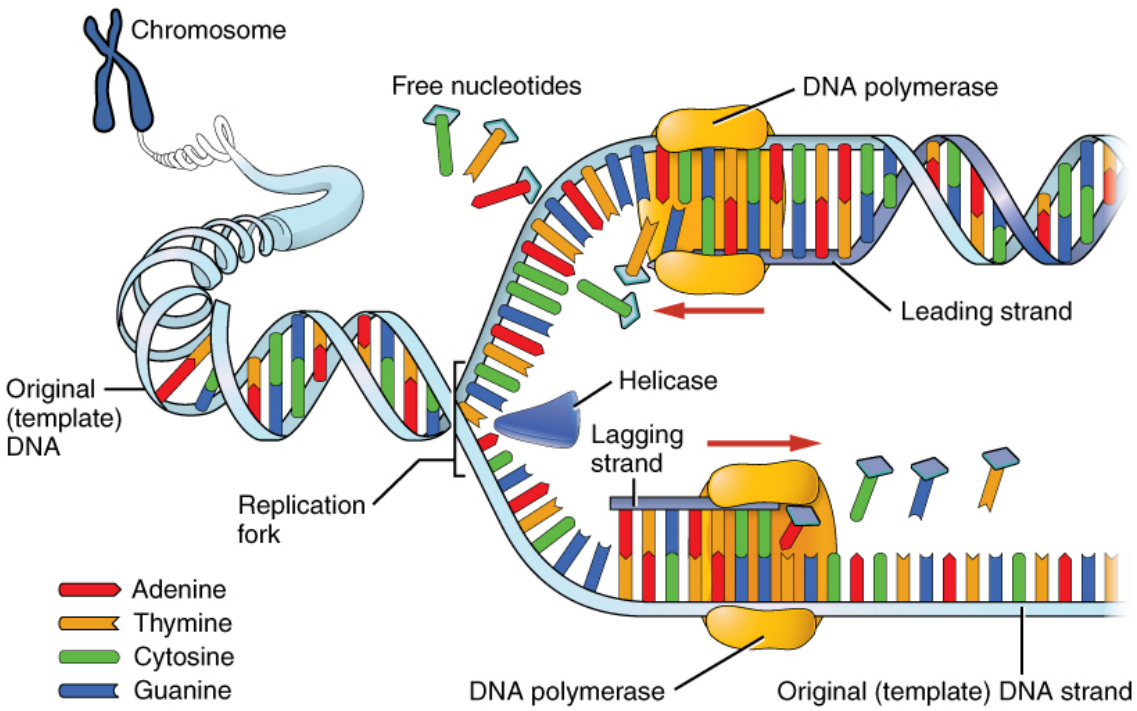

Modified ddNTP

- Modified ddNTP's were developed with a missing $3'$ hydroxyl group which prevented the chain from developing any further
- These were marked (either with radioactive phosphorous or later with a fluorescent marker)
- DNA replication was then carried out with normal ddNTP's and a weak concentration of these modified ddNTP's
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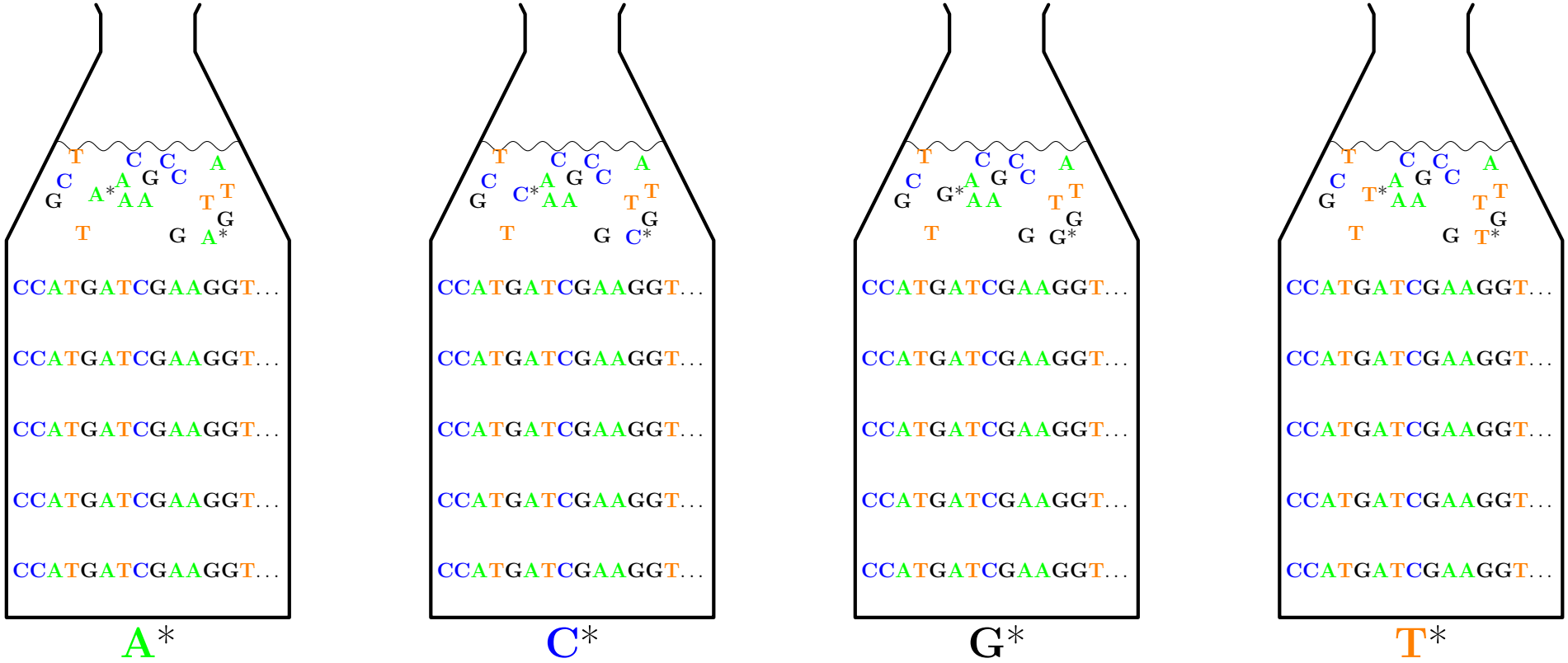

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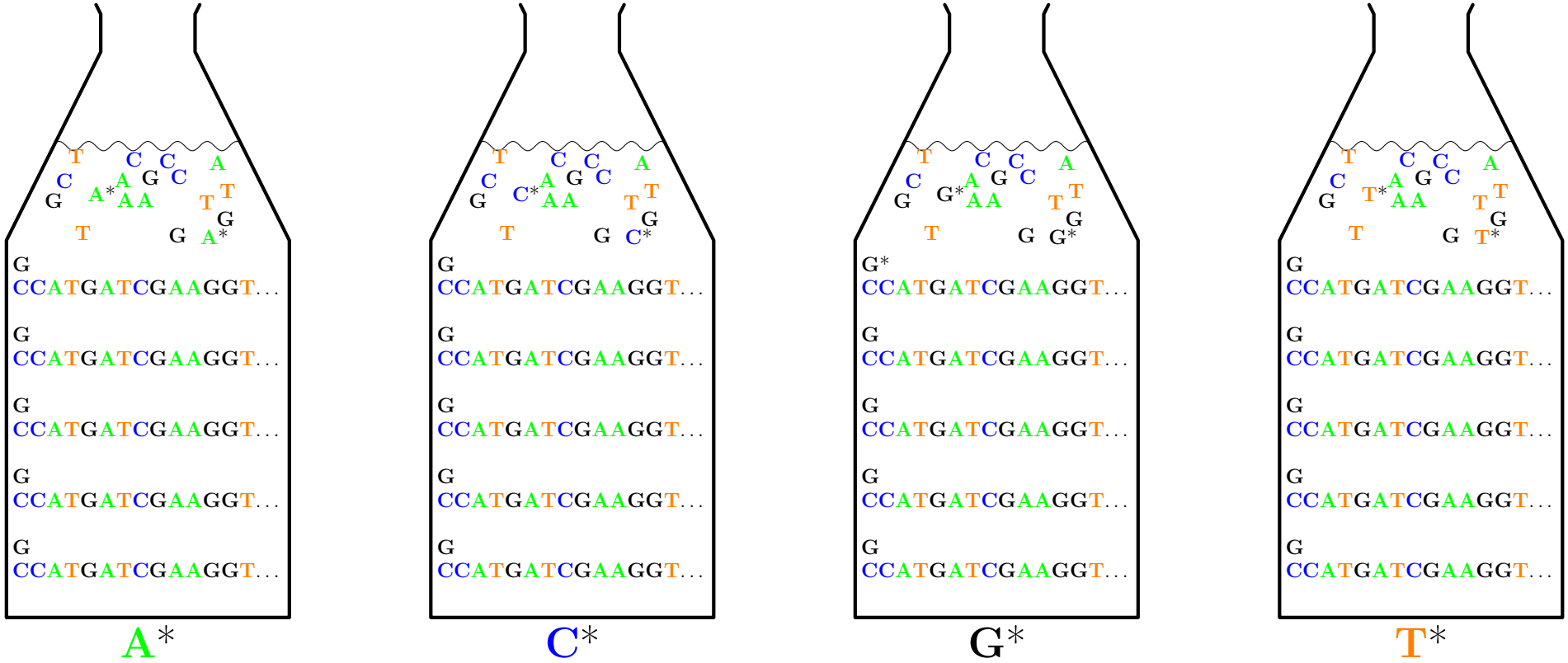

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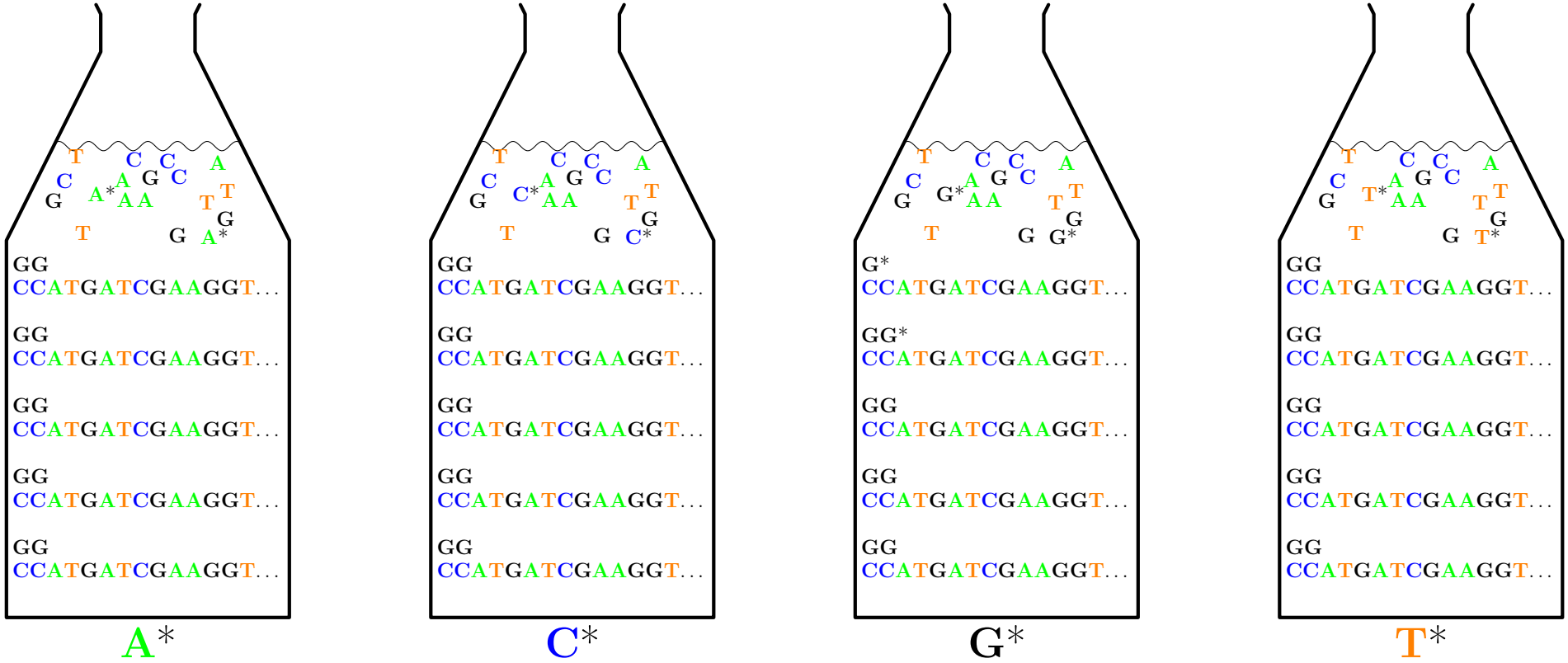

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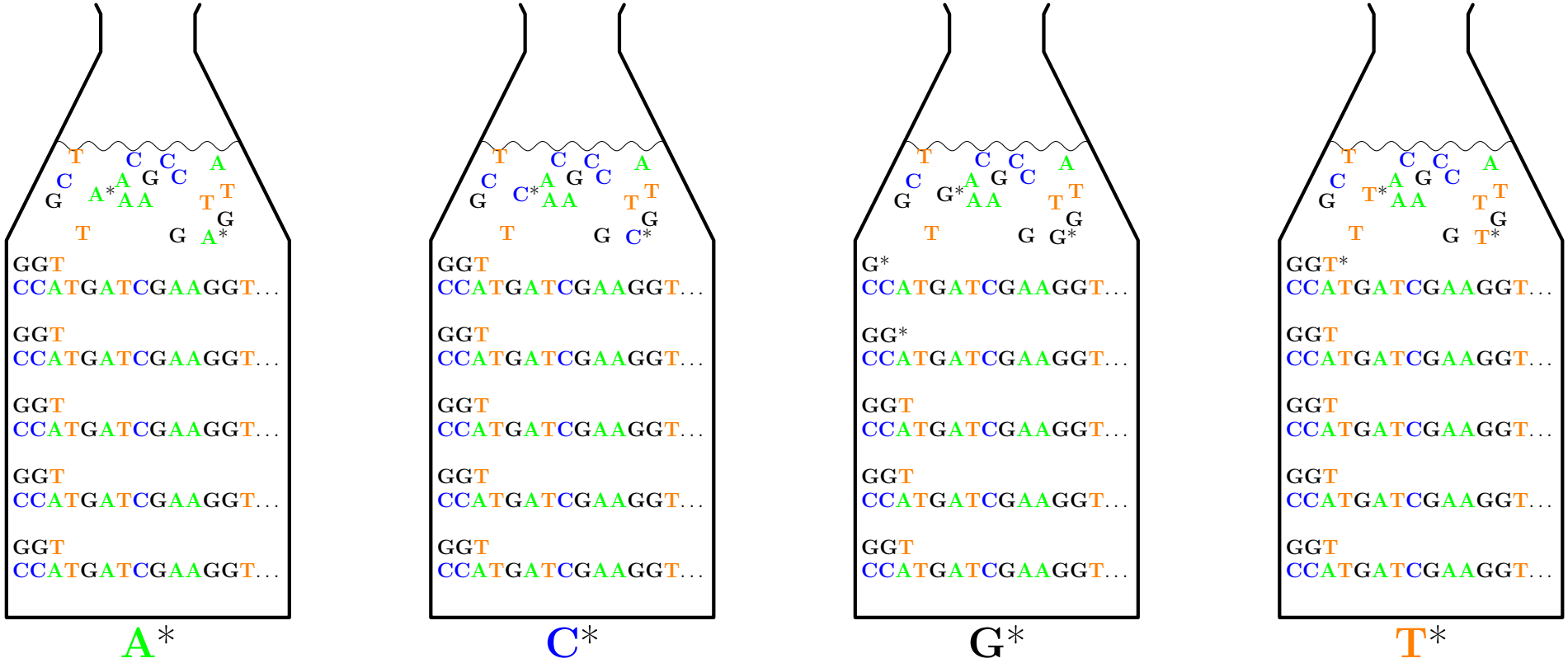

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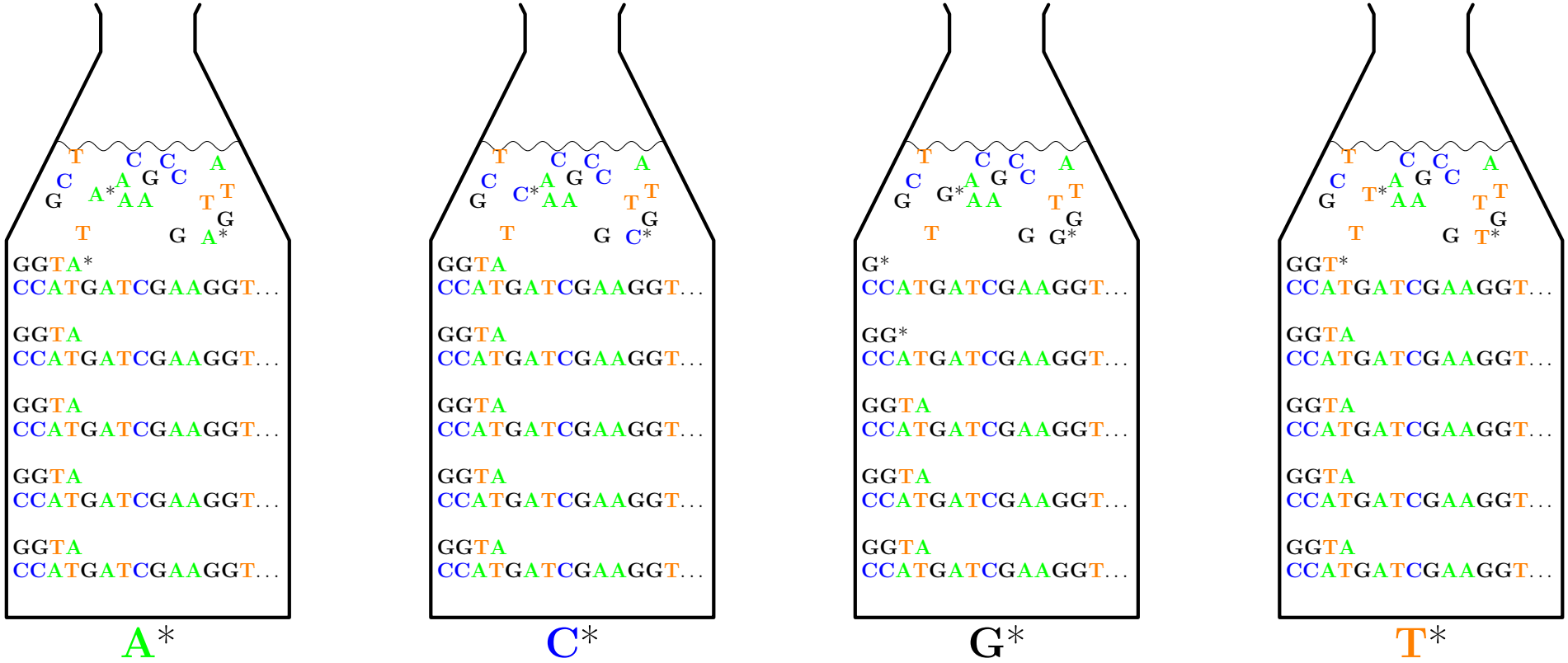

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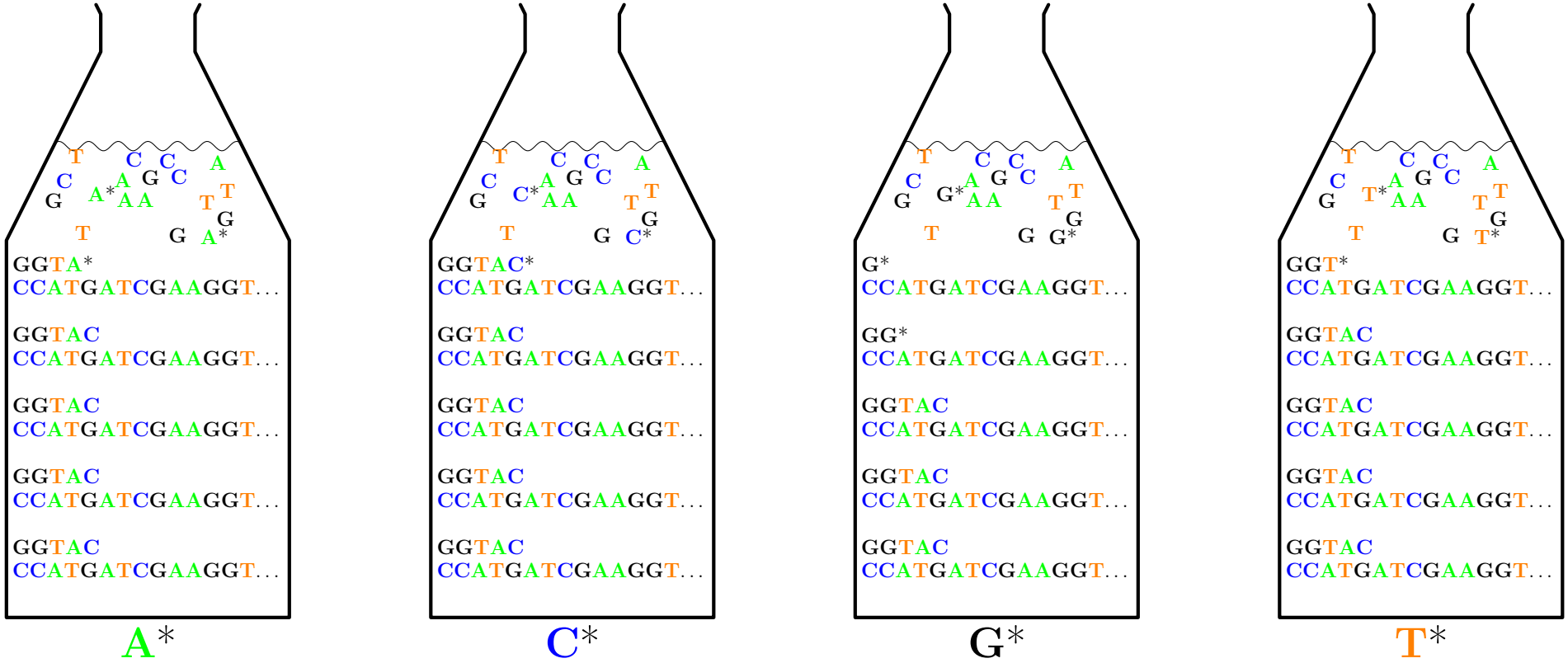

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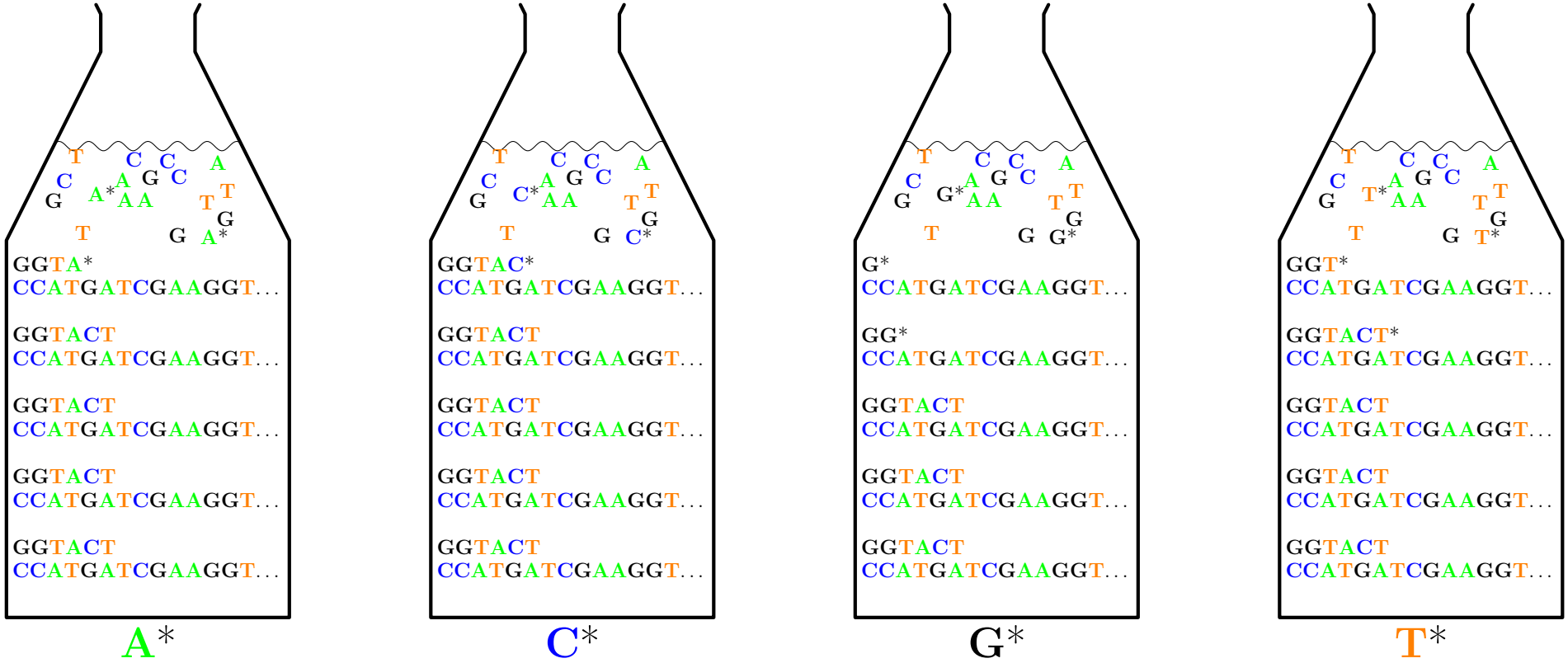

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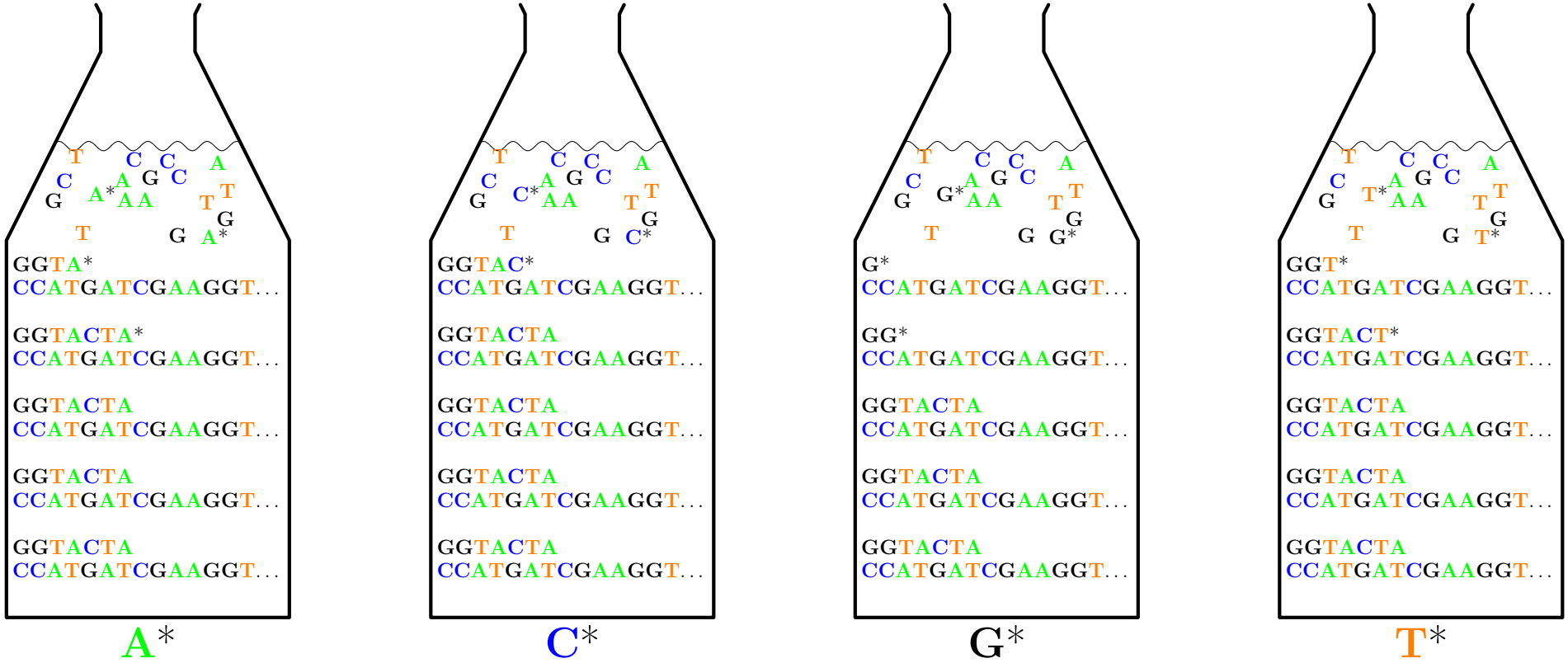

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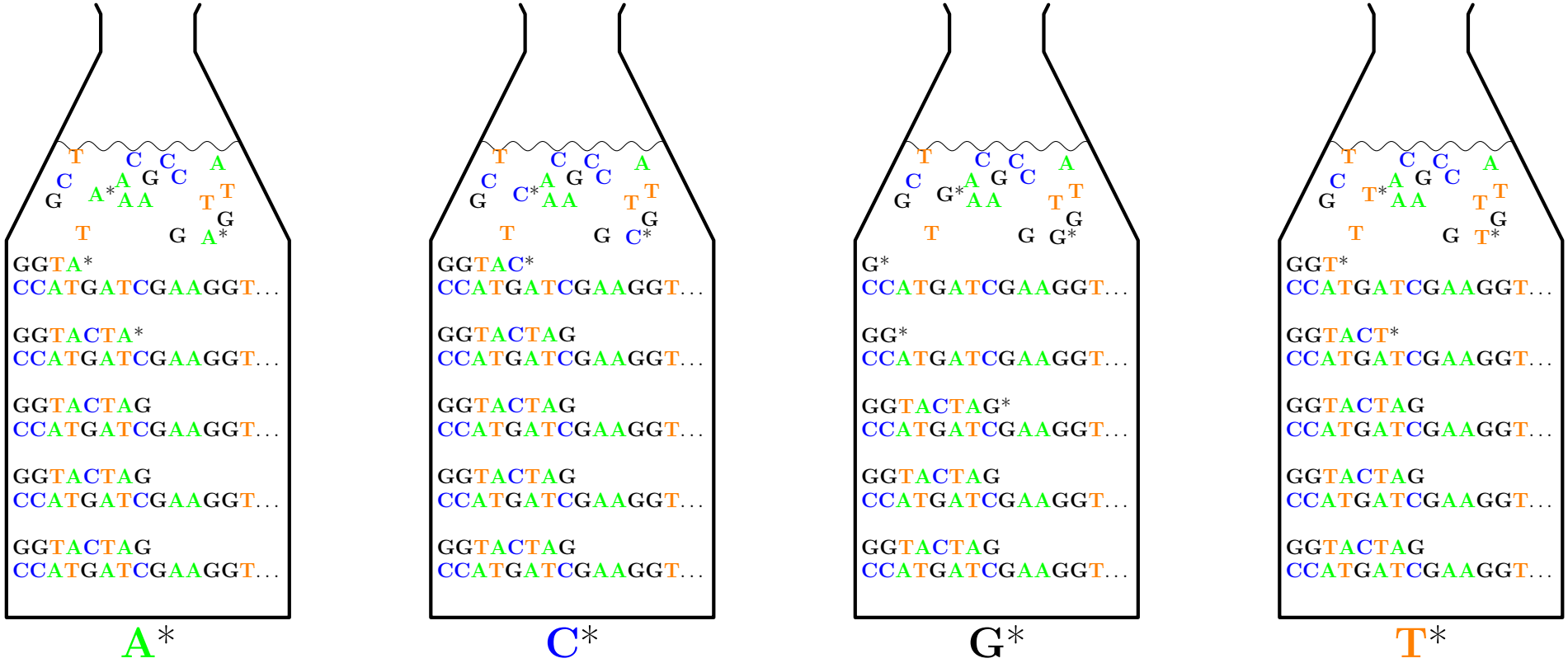

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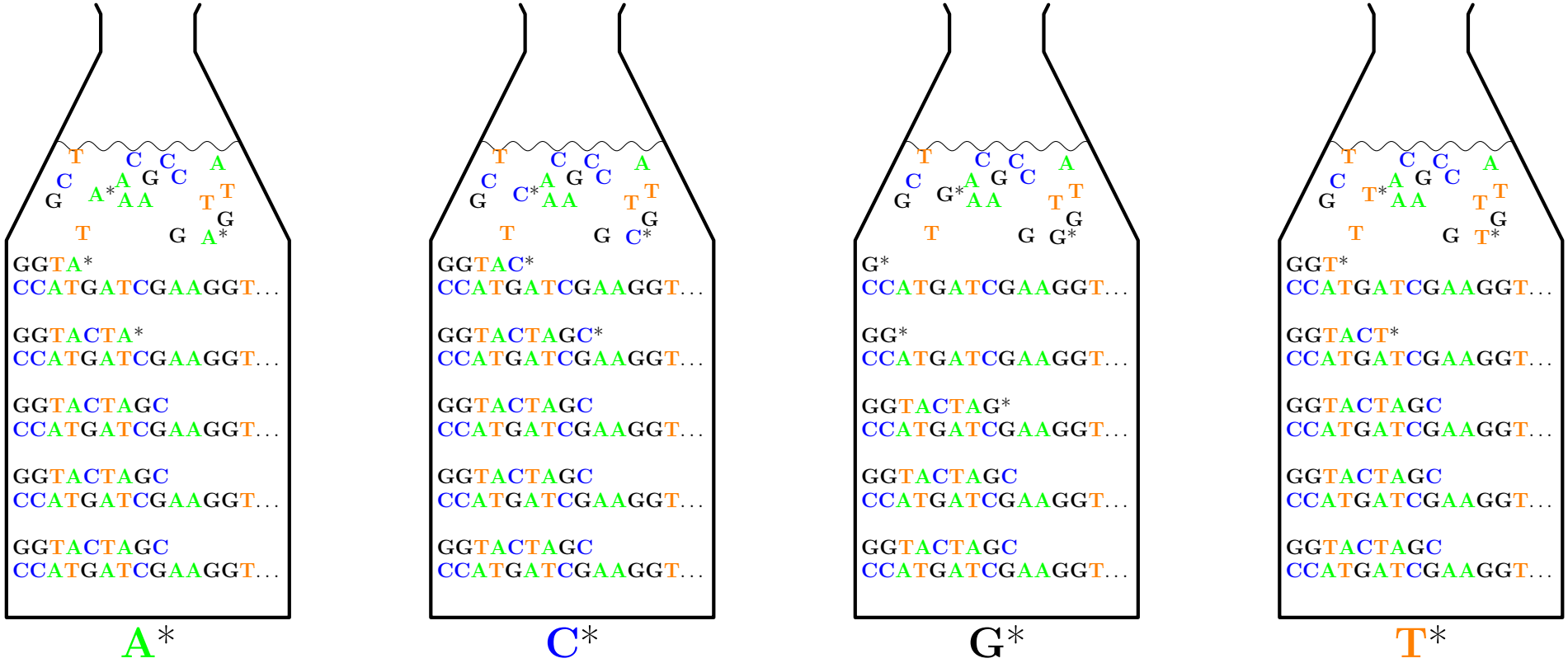

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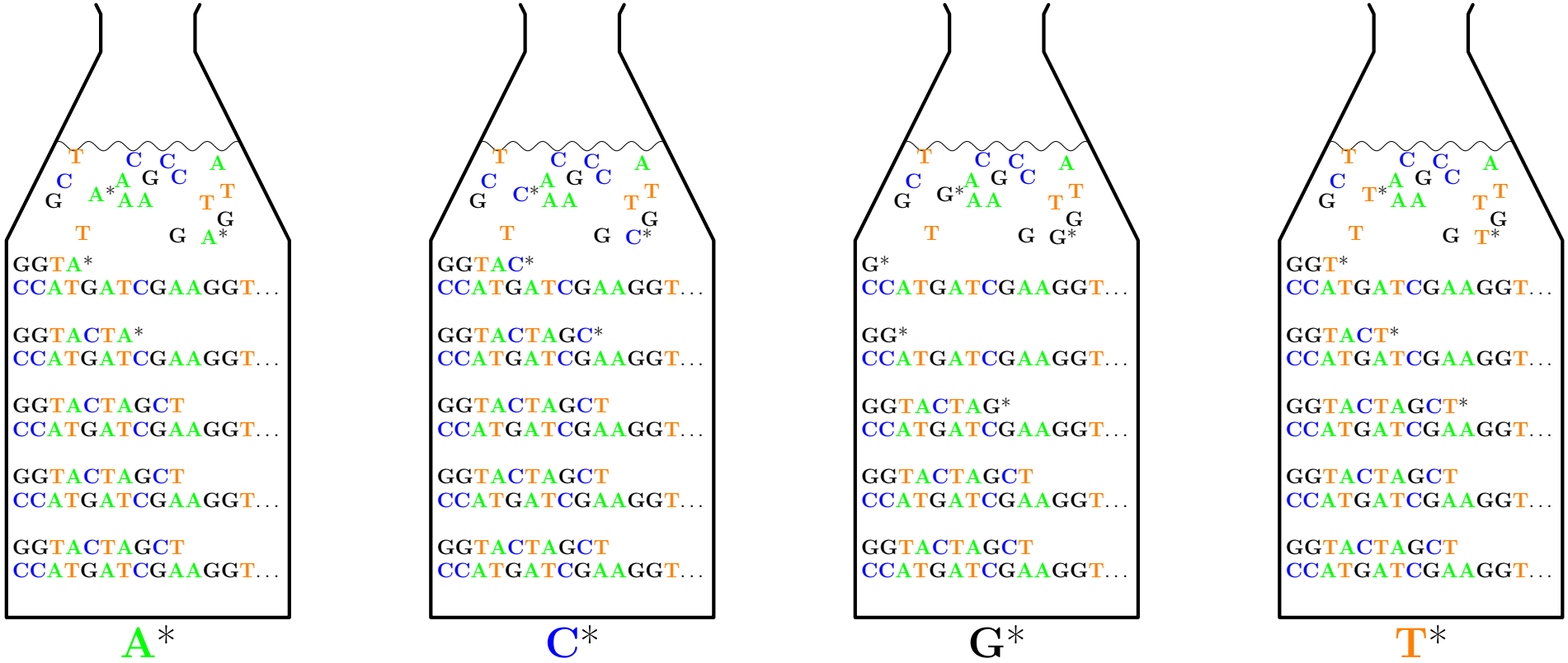

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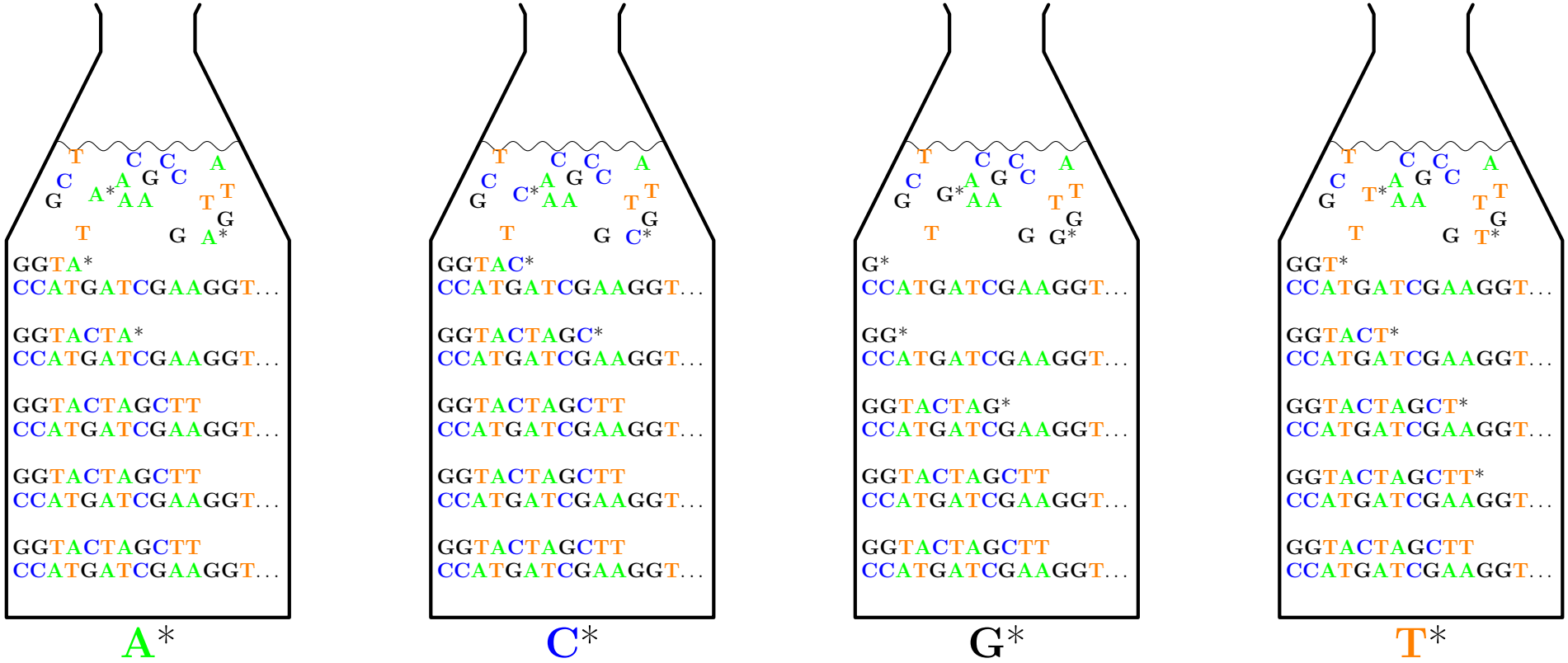

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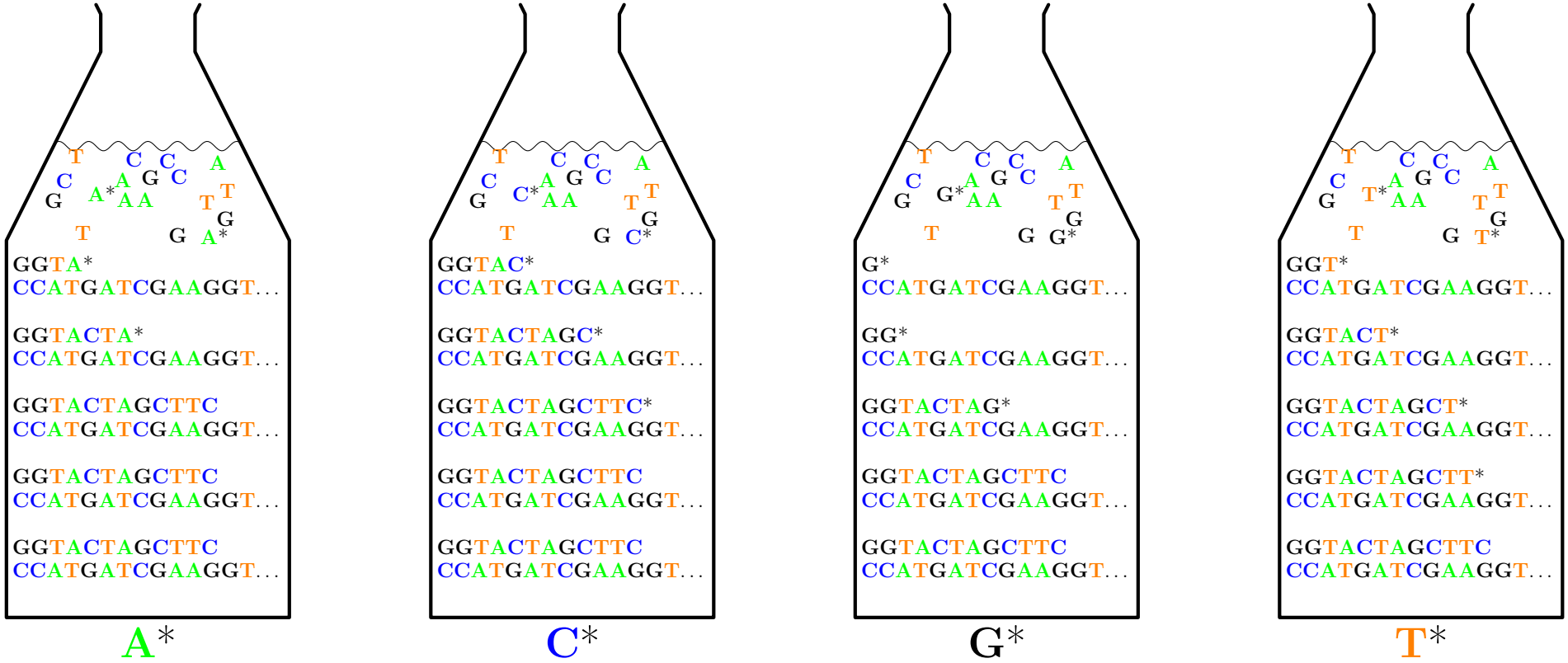

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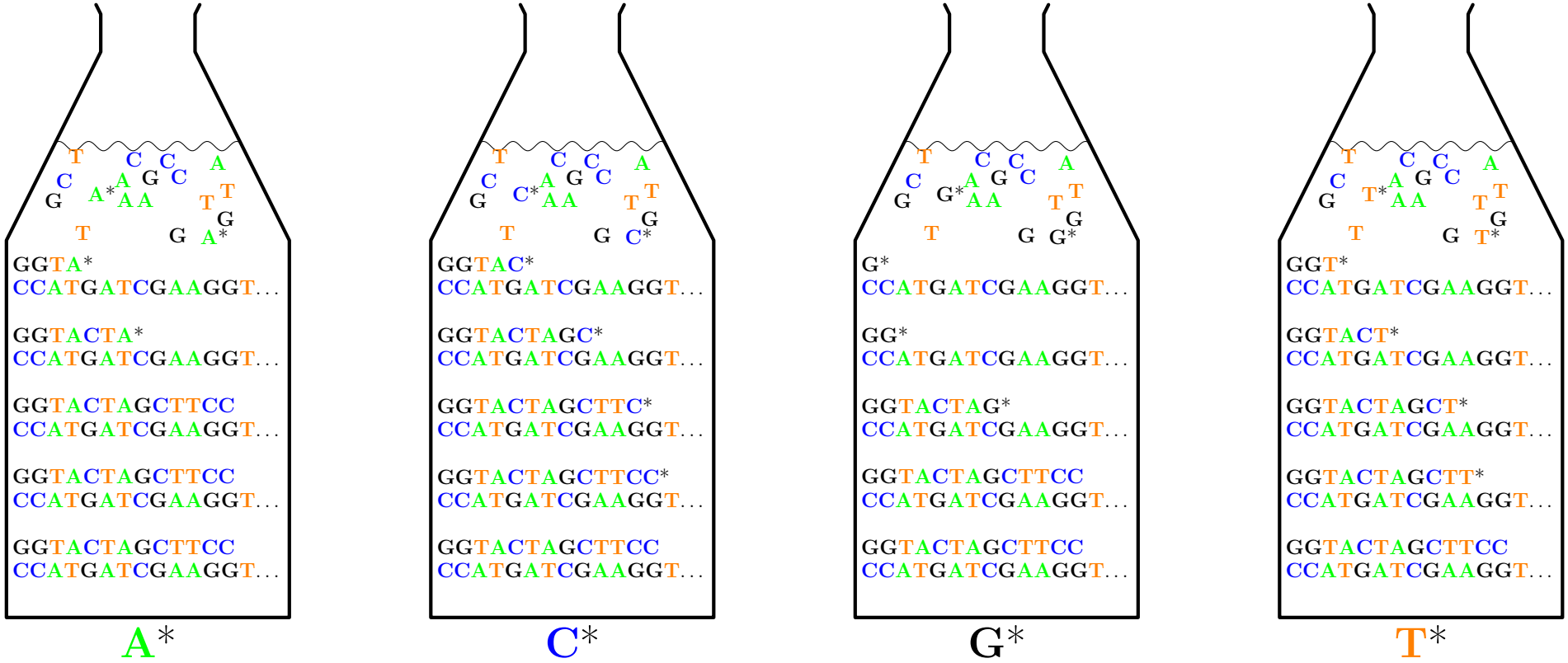

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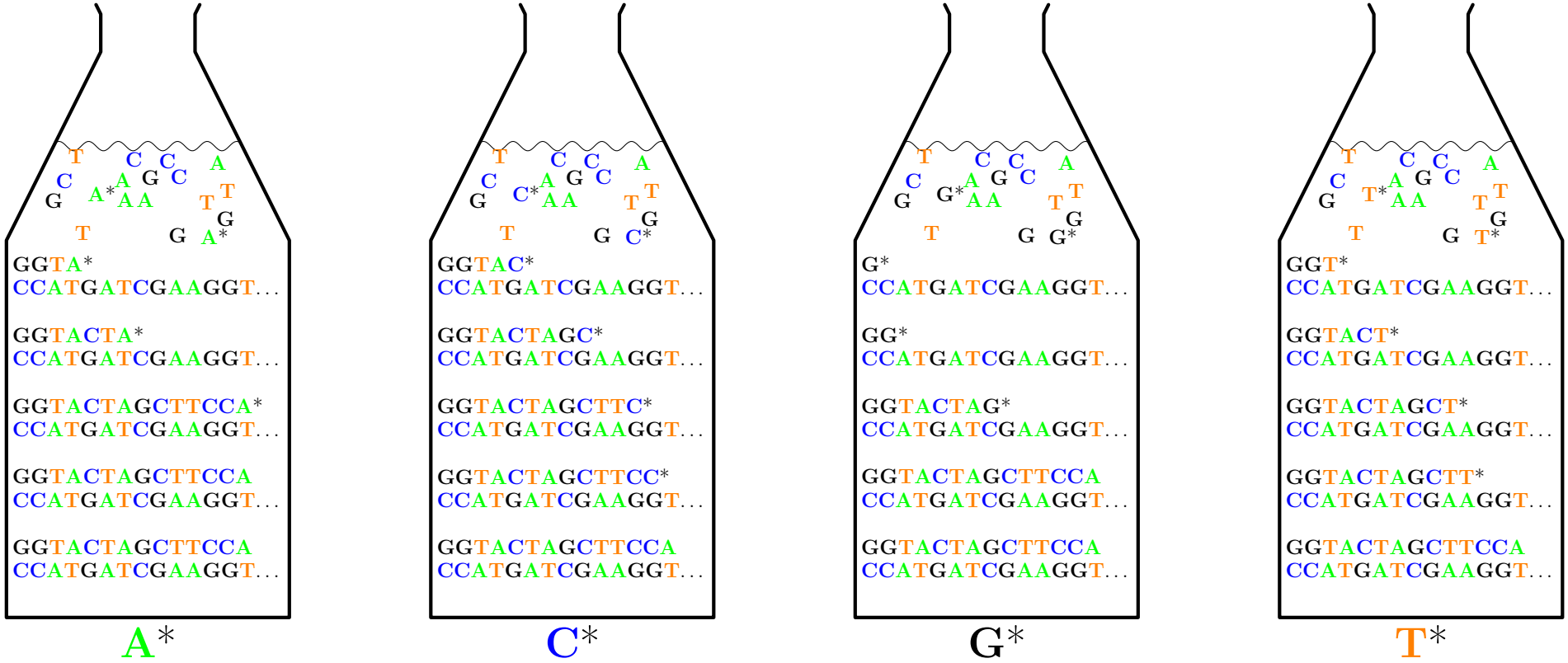

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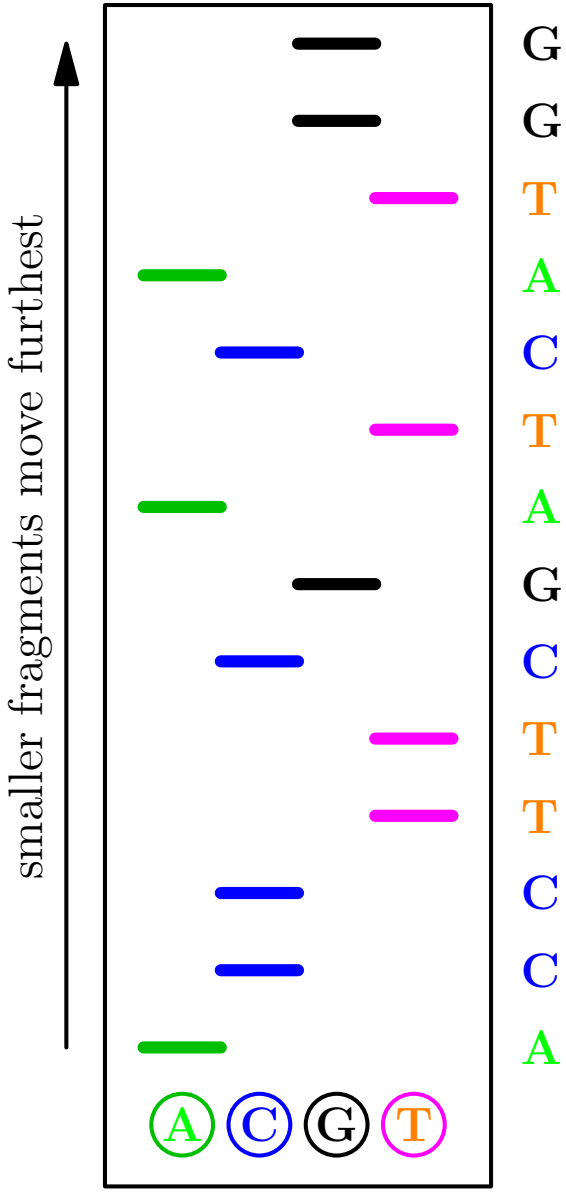

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- From the distance travelled and the colour of the last nucleotide base we can read off the sequence
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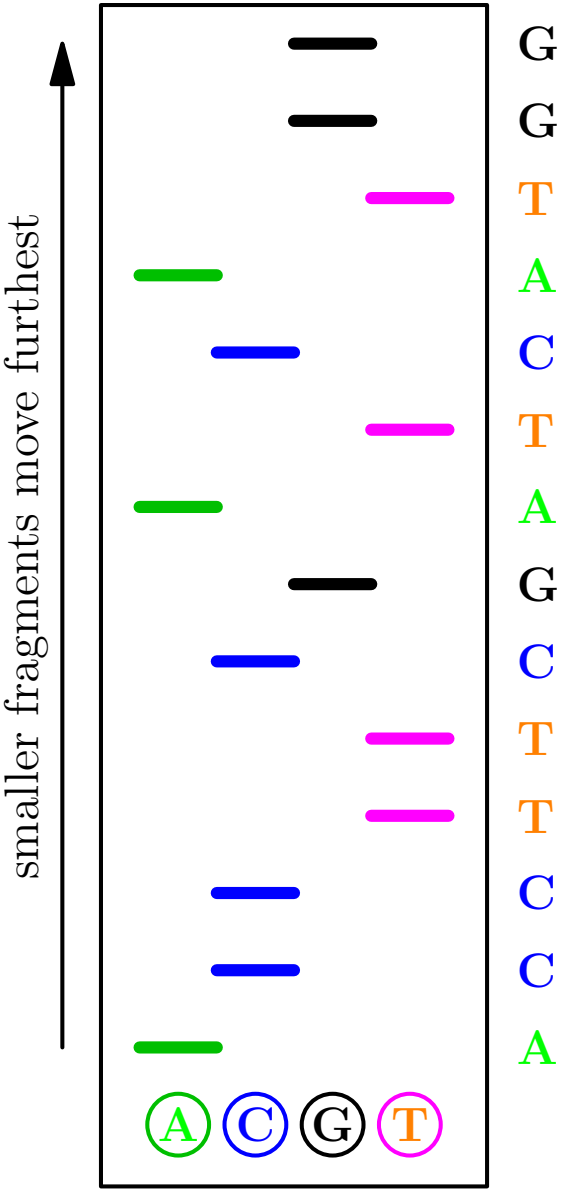

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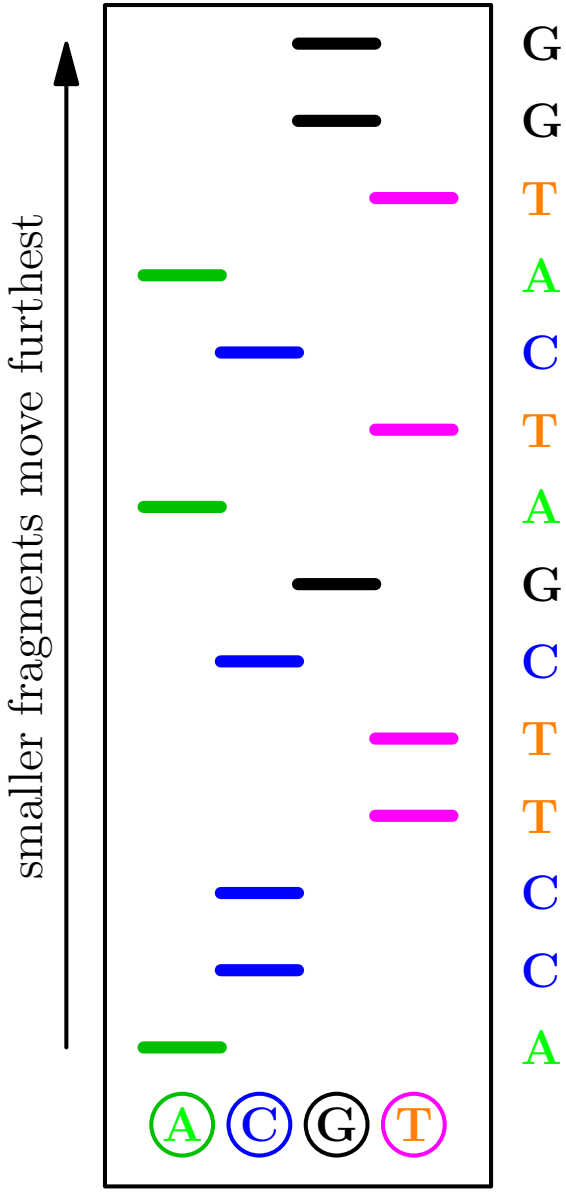

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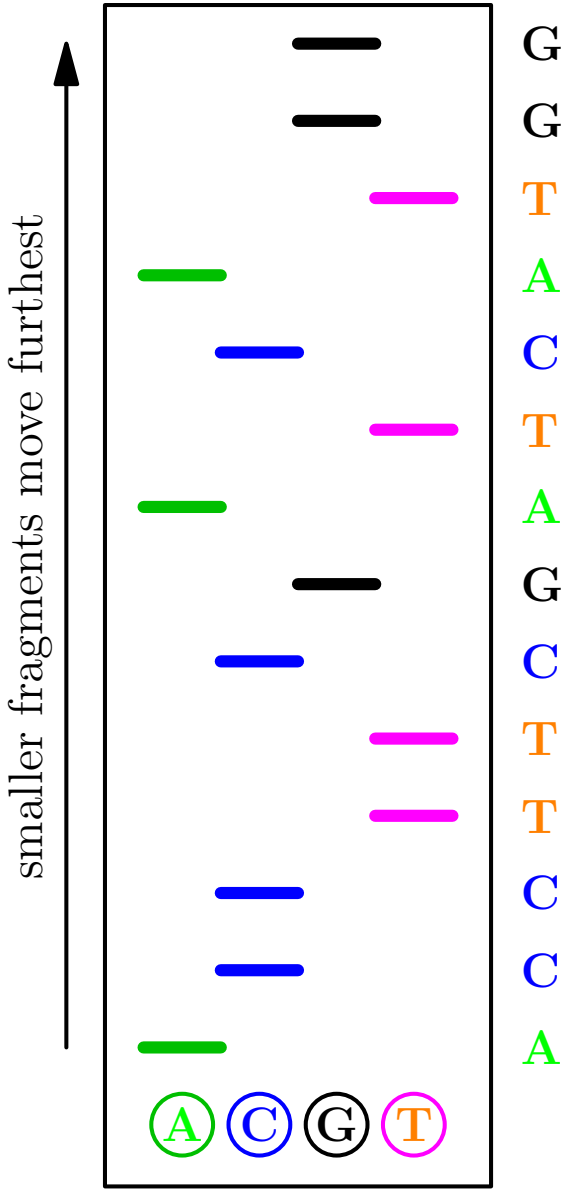

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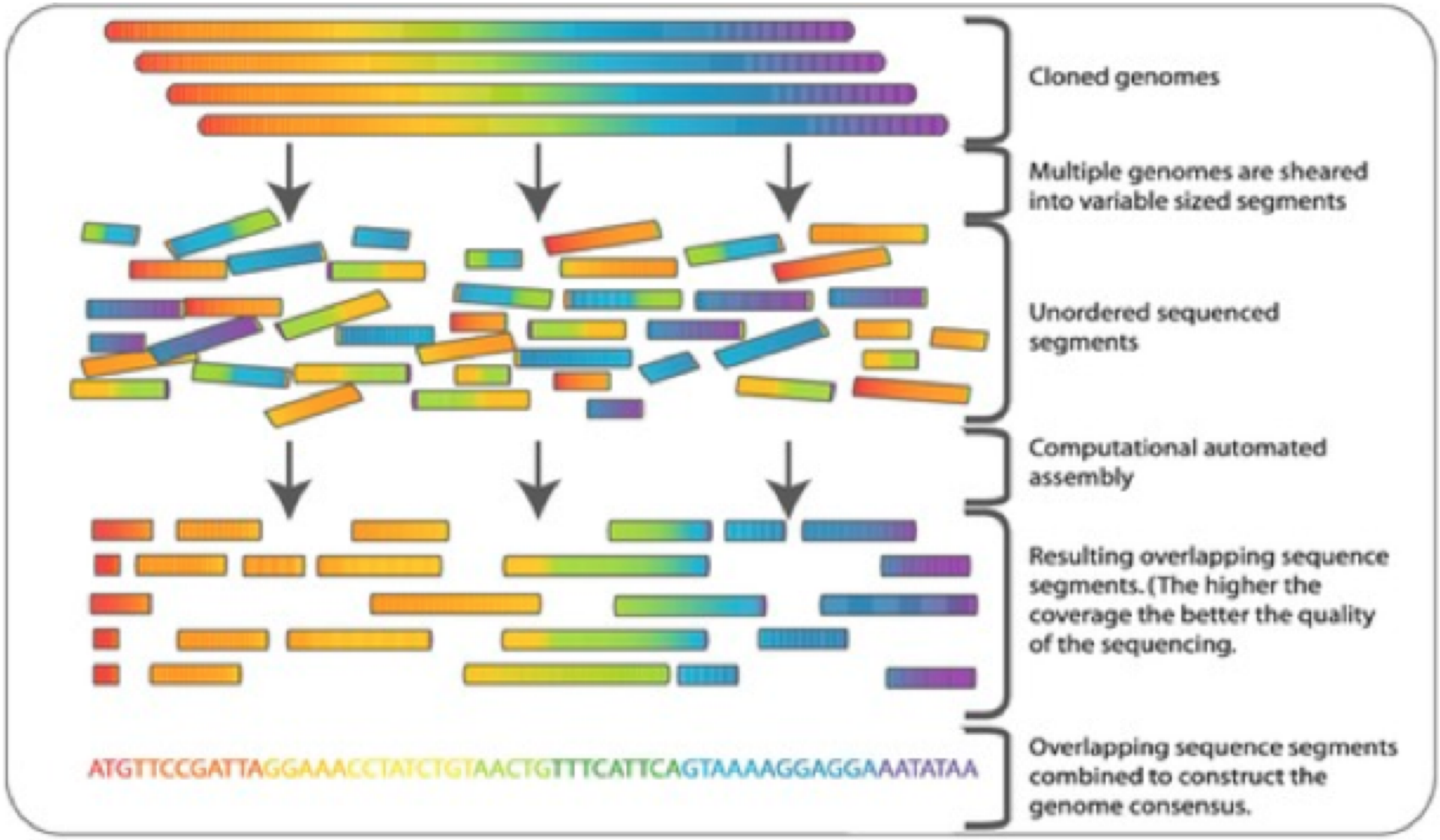
Shotgun Sequencing

To save careful preparation researcher developed shotgun sequencing



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- Sanger sequences only allowed you to sequence "reads" of around 1000bp
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- You can sequence both sides of a large fragment so you know the approximate relationship between reads

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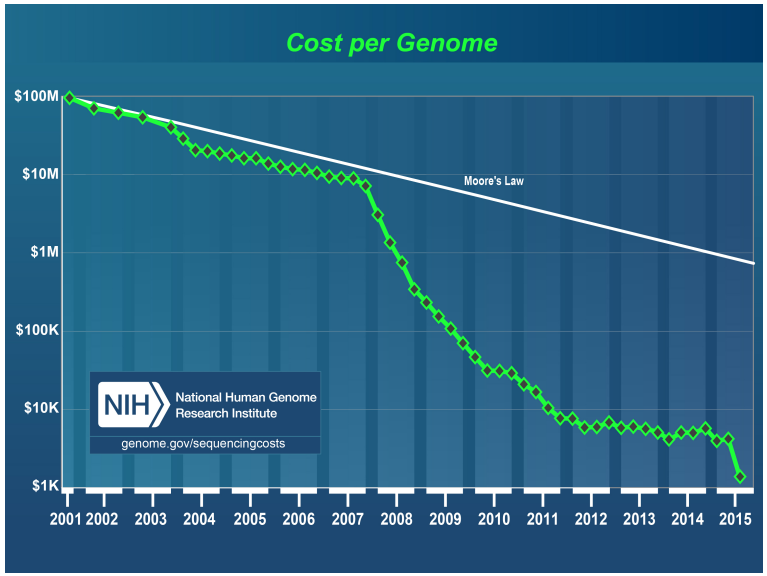

Improving on Sanger

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- The ideas was to use massive parallelisation
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Sequencing Techniques

- Sanger sequencing used sequencing-by-synthesis
  - ⋆ Using DNA-polymerase a strand of DNA is replicated
  - ⋆ It turns out this technique that ultimately triumphant
- Another approach is sequencing-by-ligation (or hybridisation)
  - ⋆ Create lots of short sections of DNA (oligonucleotides) that will stick to an unbound stretch of DNA (or RNA)
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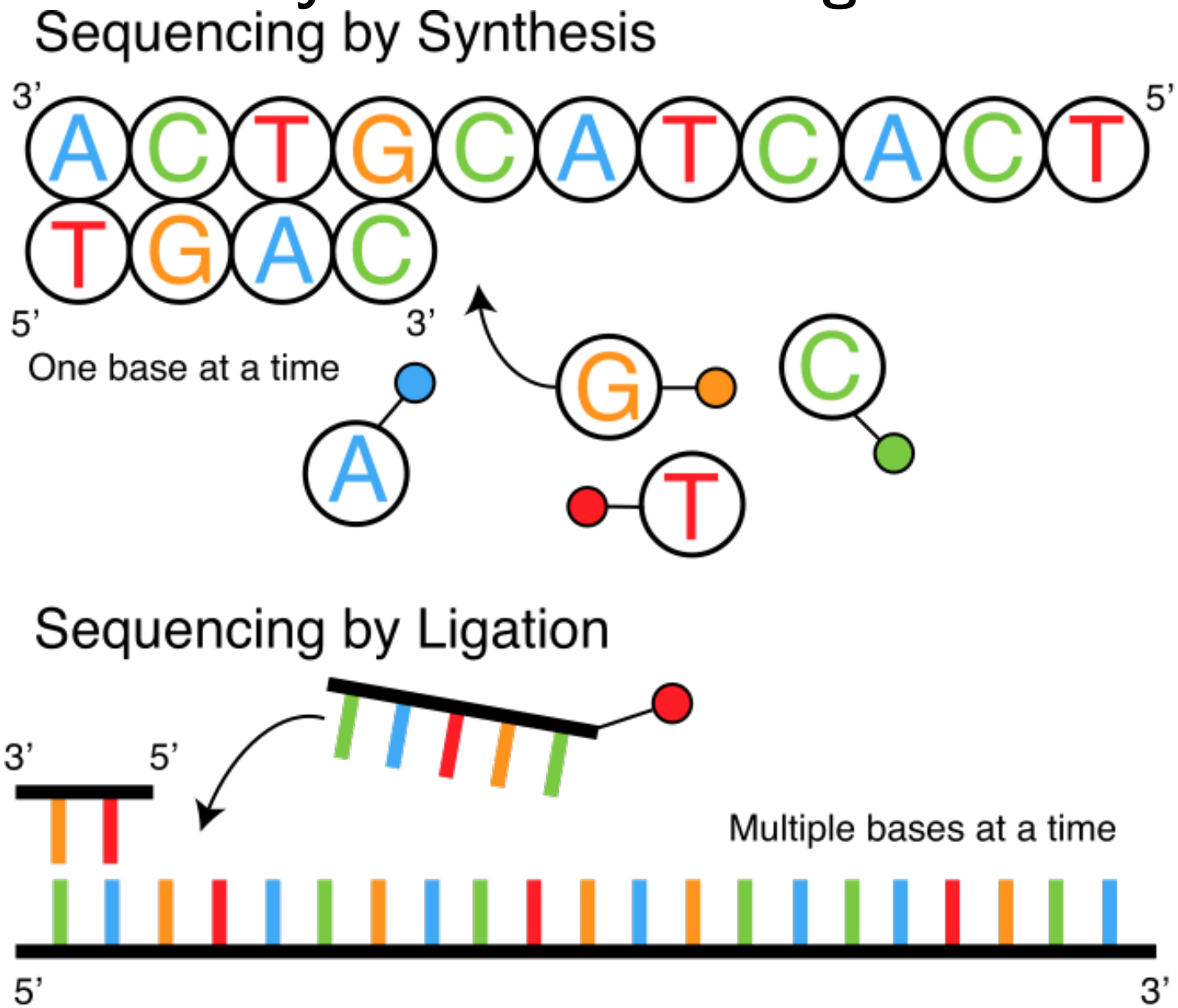

454 Sequencer

- The first high throughput sequencer was the Roche 454 machine
- It amplified fragments of DNA attached to a microbead
- These are contained in micro-wells
- Sequencing-by-synthesis is carried out
- Added one type of base at a time with fluorescent label
- Can have difficulty distinguishing regions with repeated base

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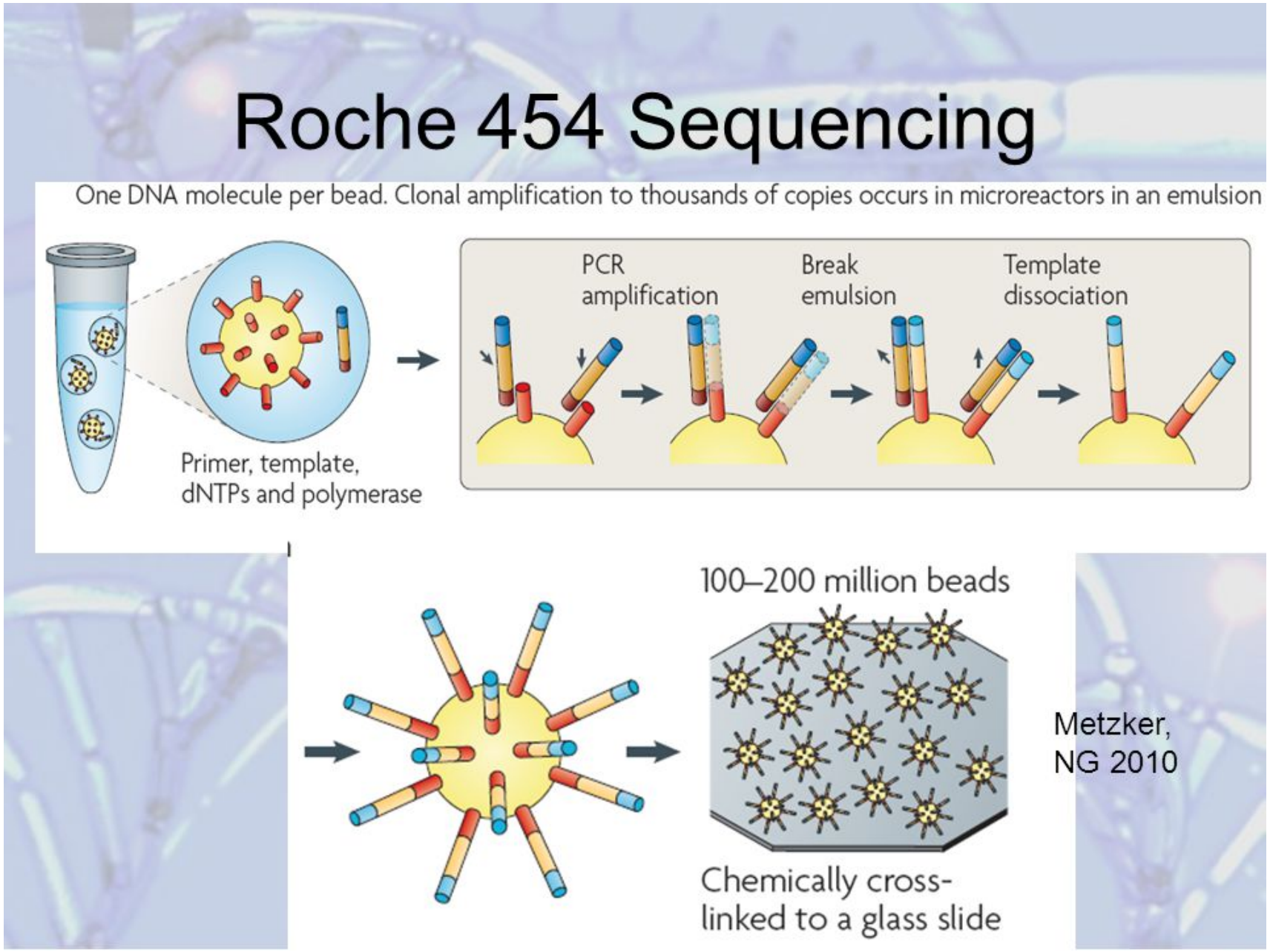
Water in Oil Emulsion PCR



Illumina

- The dominant next generation machines are produced by Illumina (formally Solexa)
- It uses a technique now called bridge amplification to create clusters of identical strands on a 2-D surface (substrate)
- The machine can sequence 100's of million clusters at once
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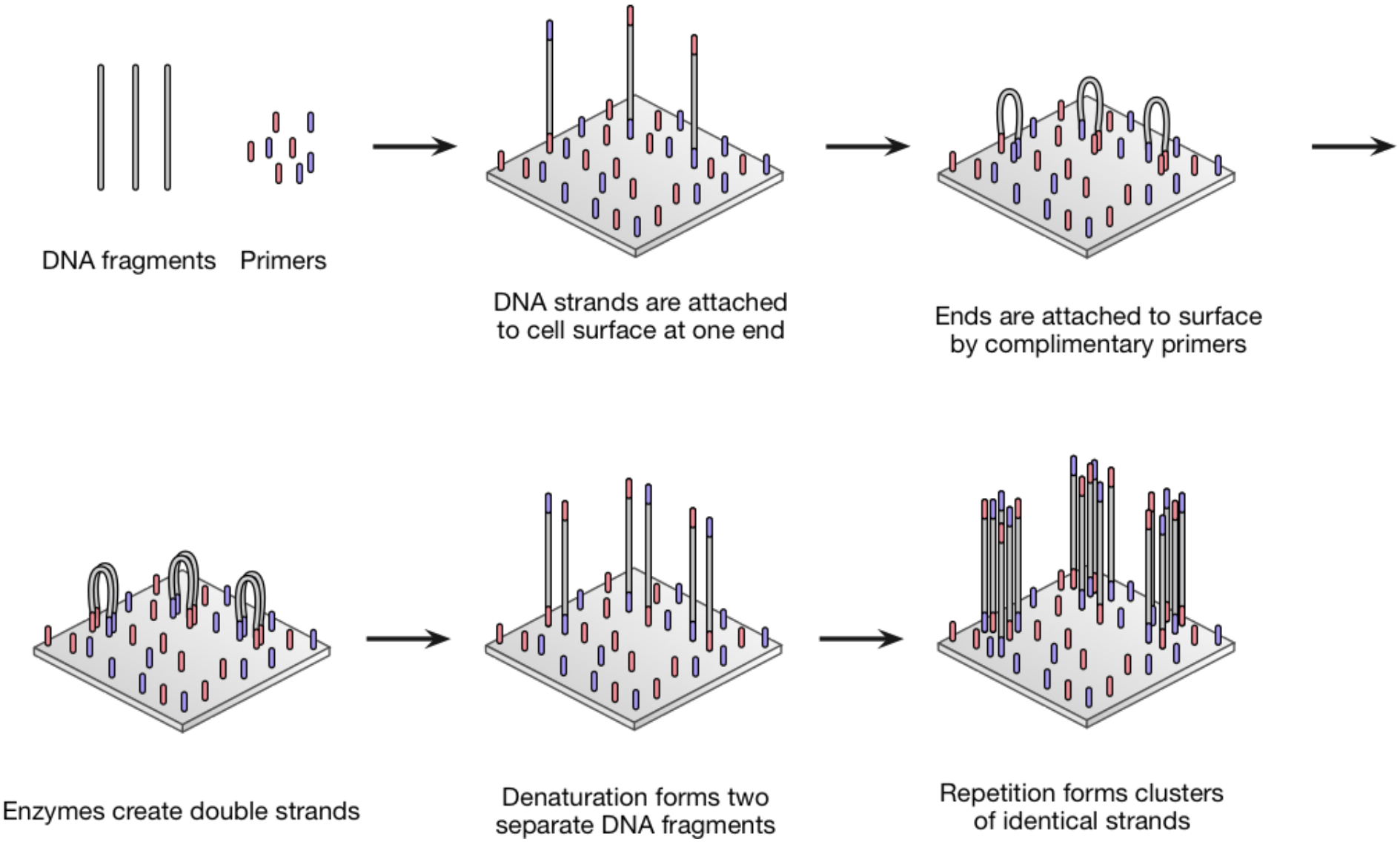
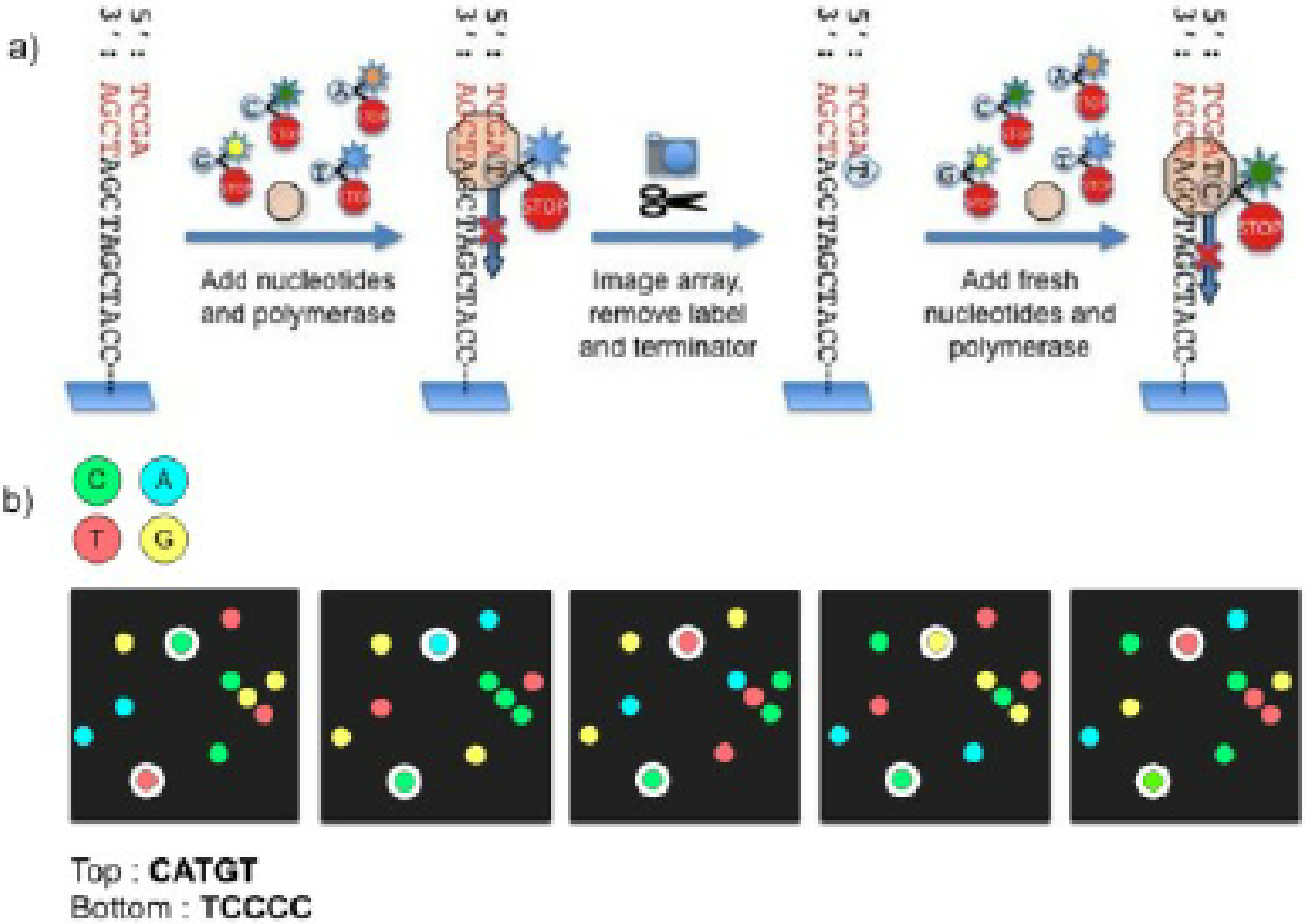
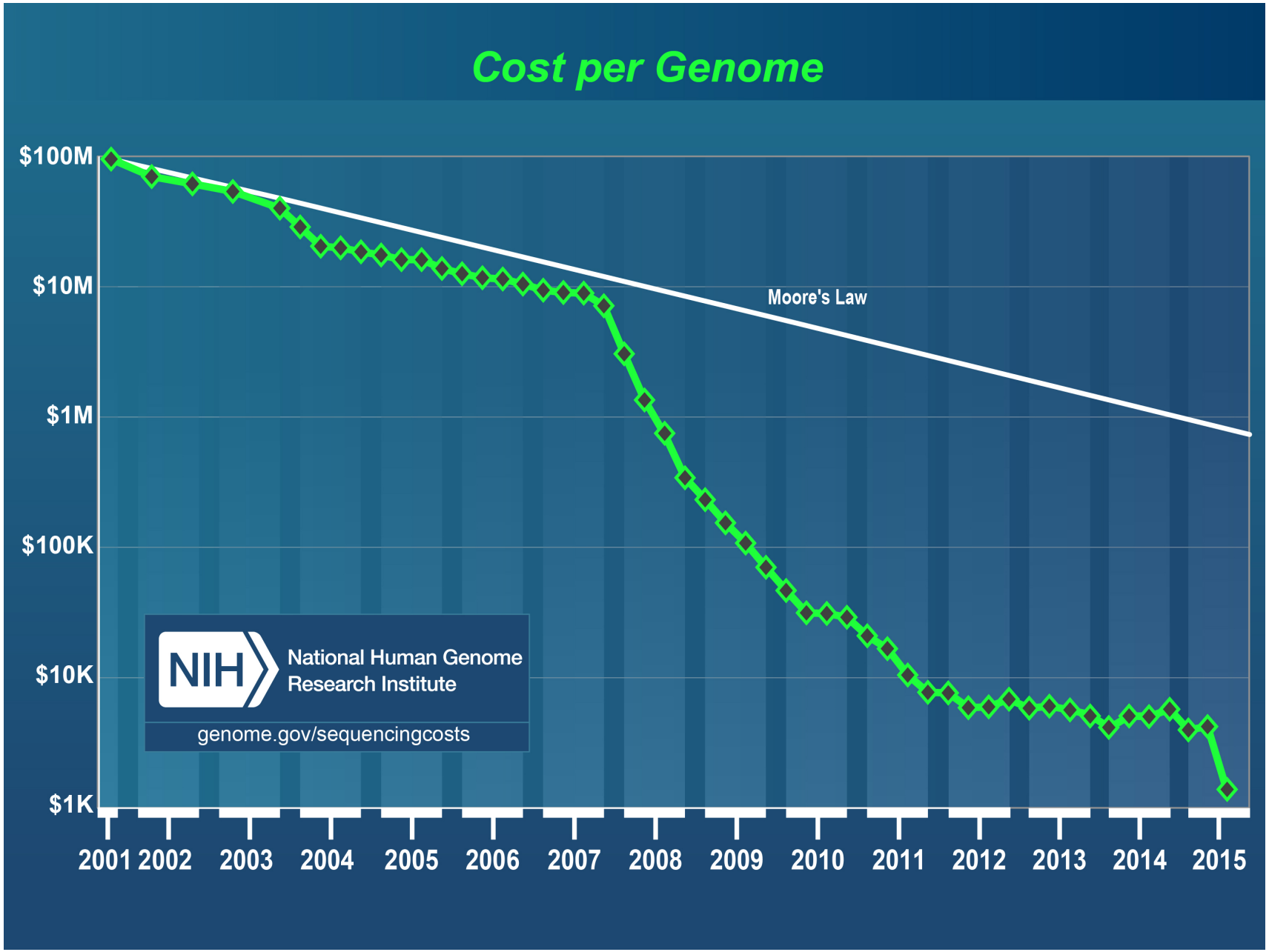


Identifying Sequence



Cost Per Human Genome



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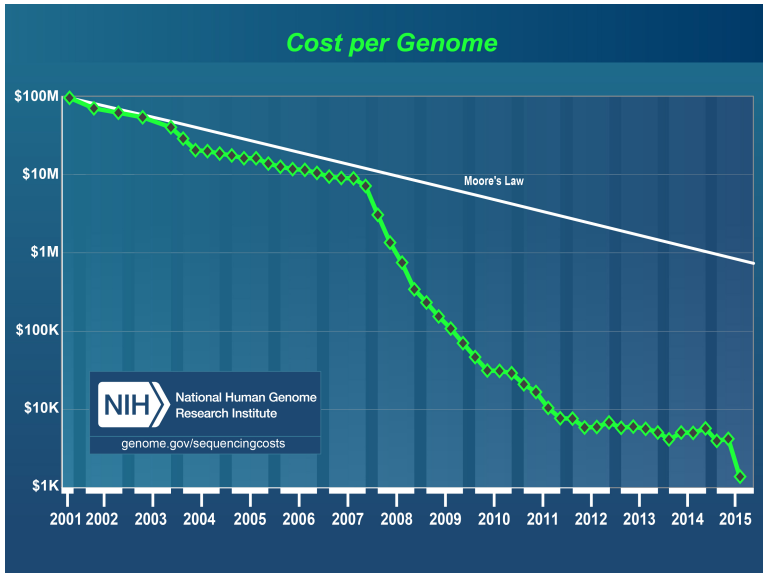
Outline

1. Sanger Sequencing
2. Next Generation Sequencers
3. **Third Generation Sequencers**



Single Molecule DNA Sequencing

- Having to replicate DNA can bring in errors and biases
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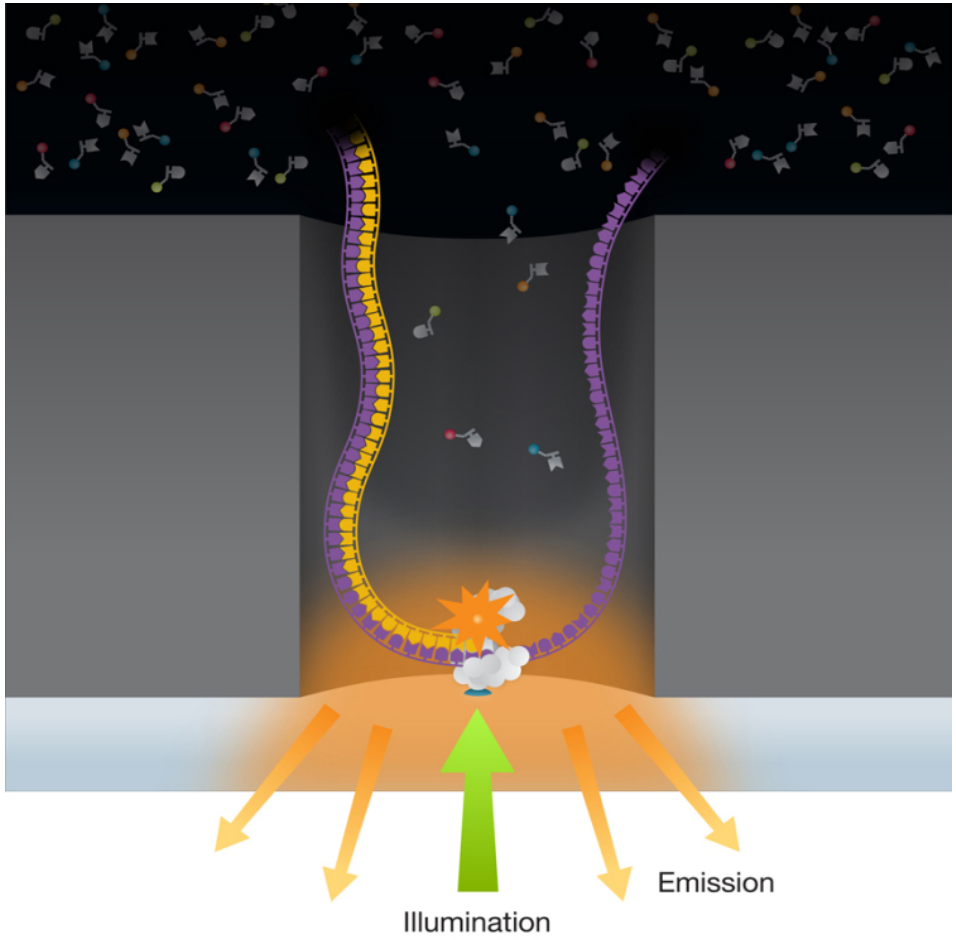

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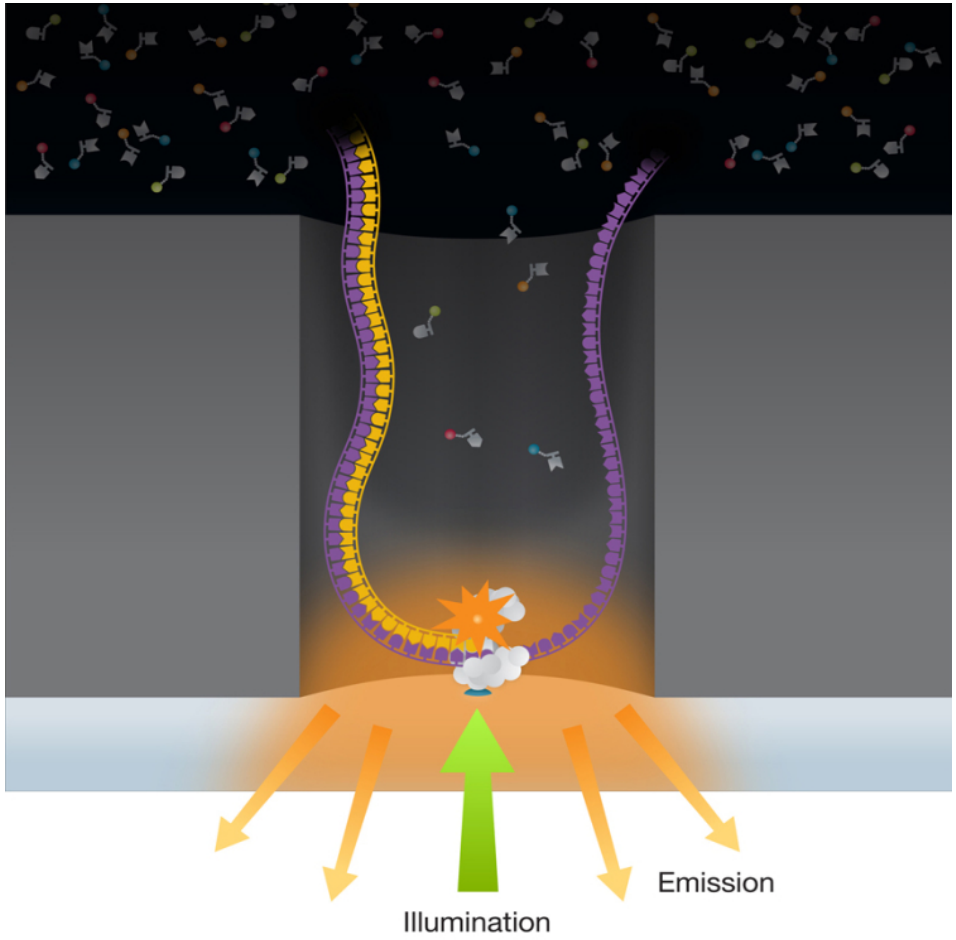

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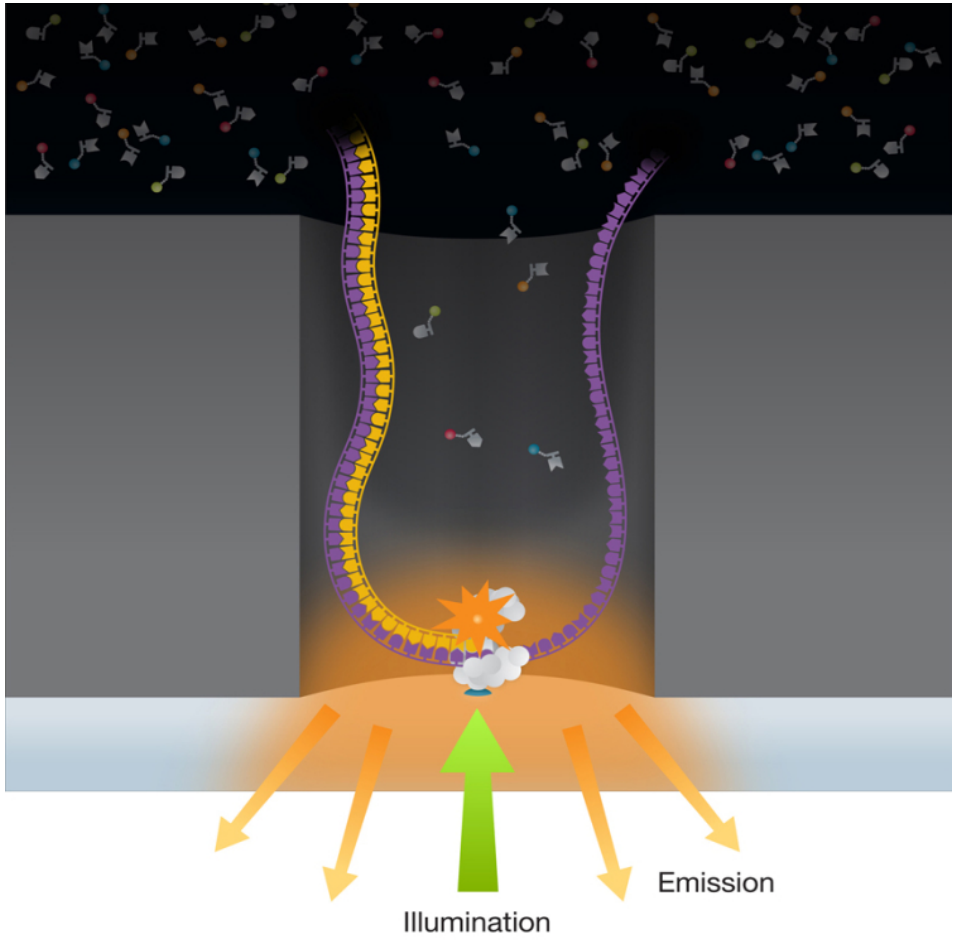

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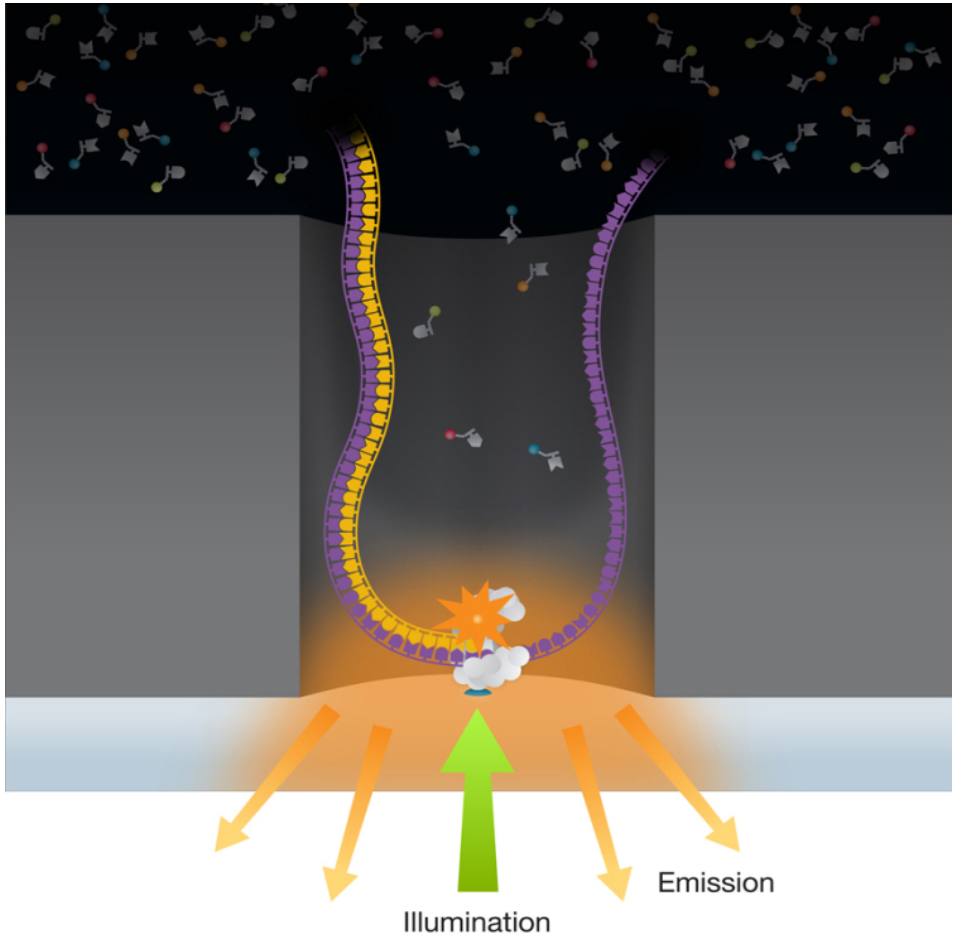

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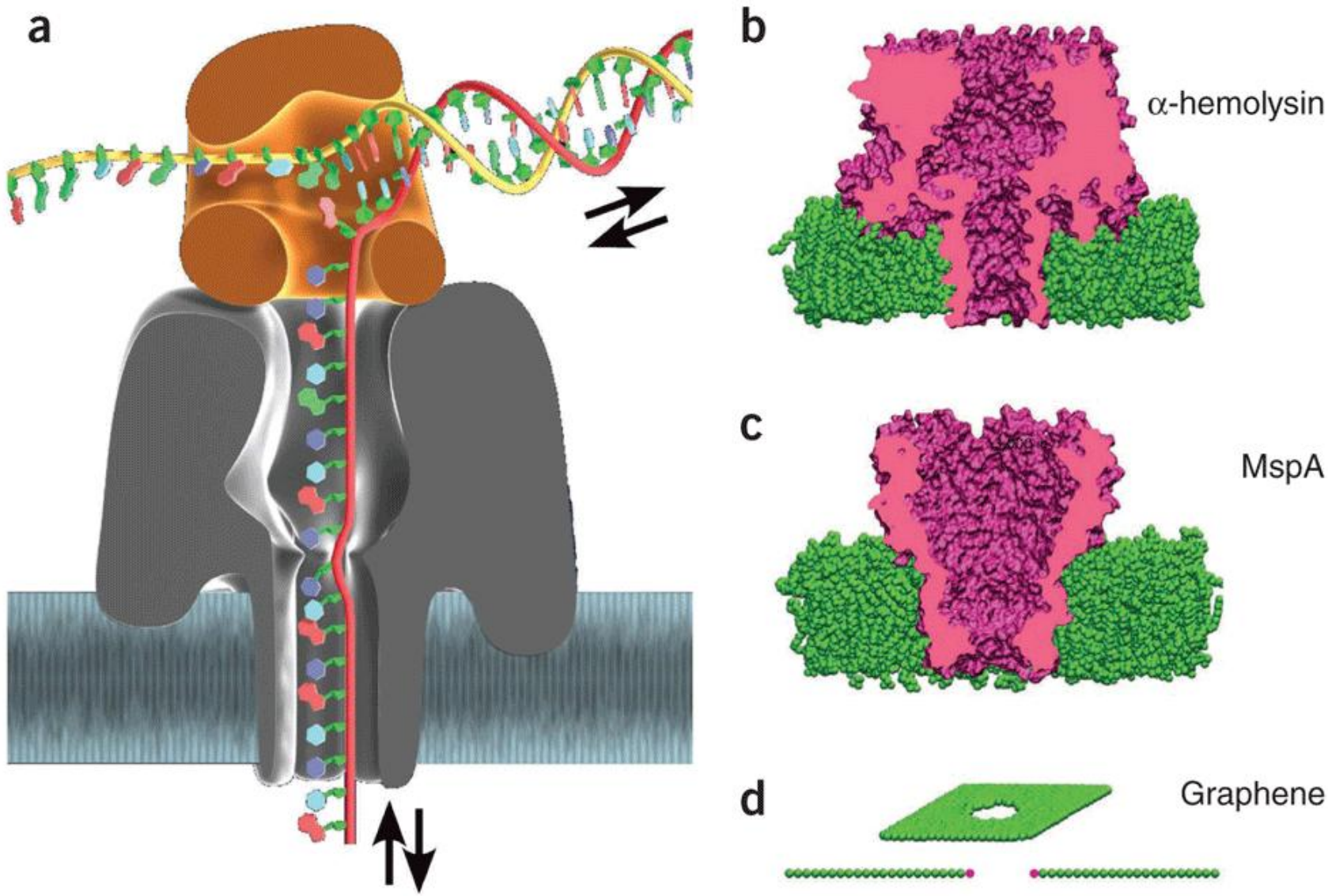

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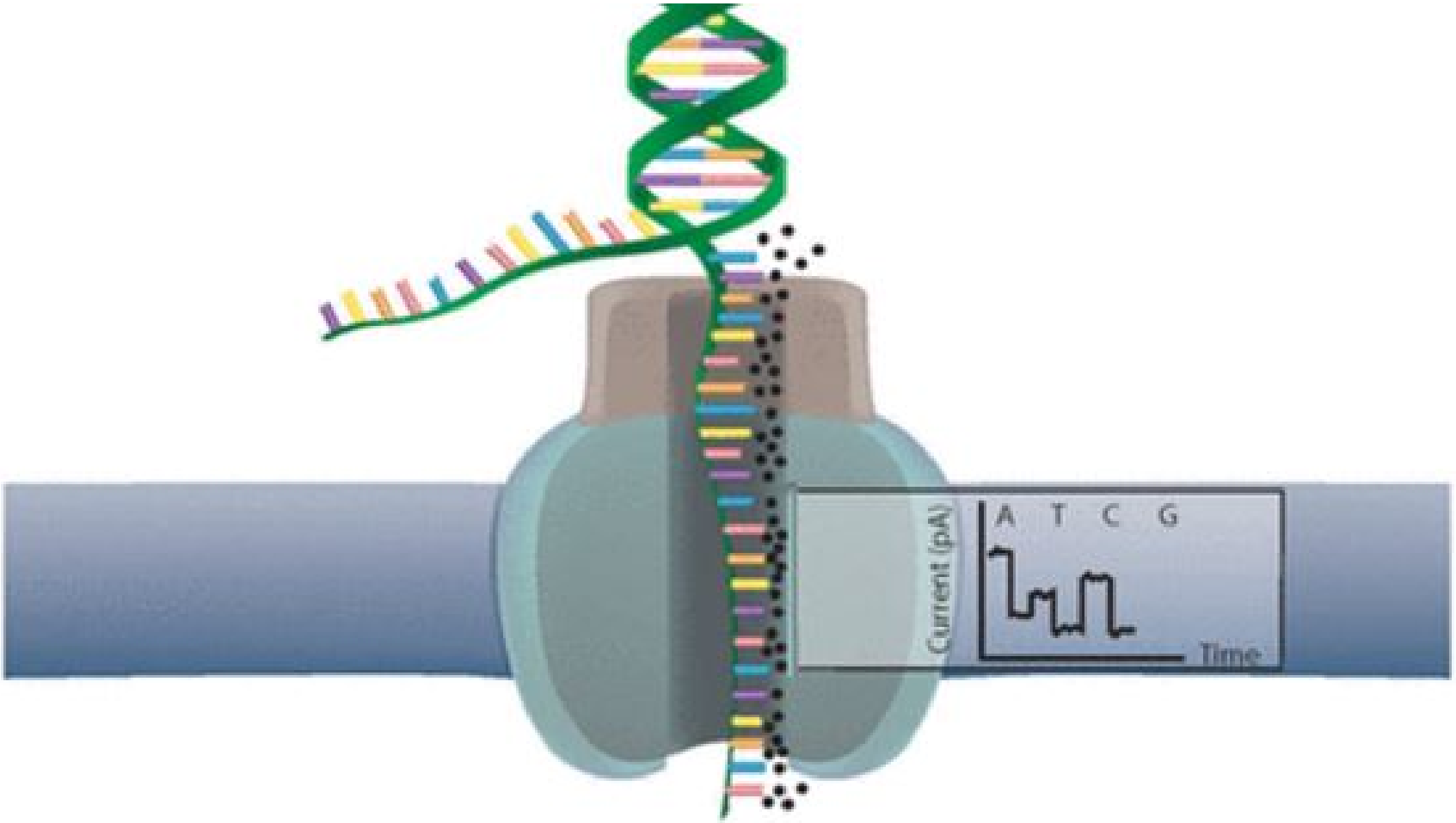

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